

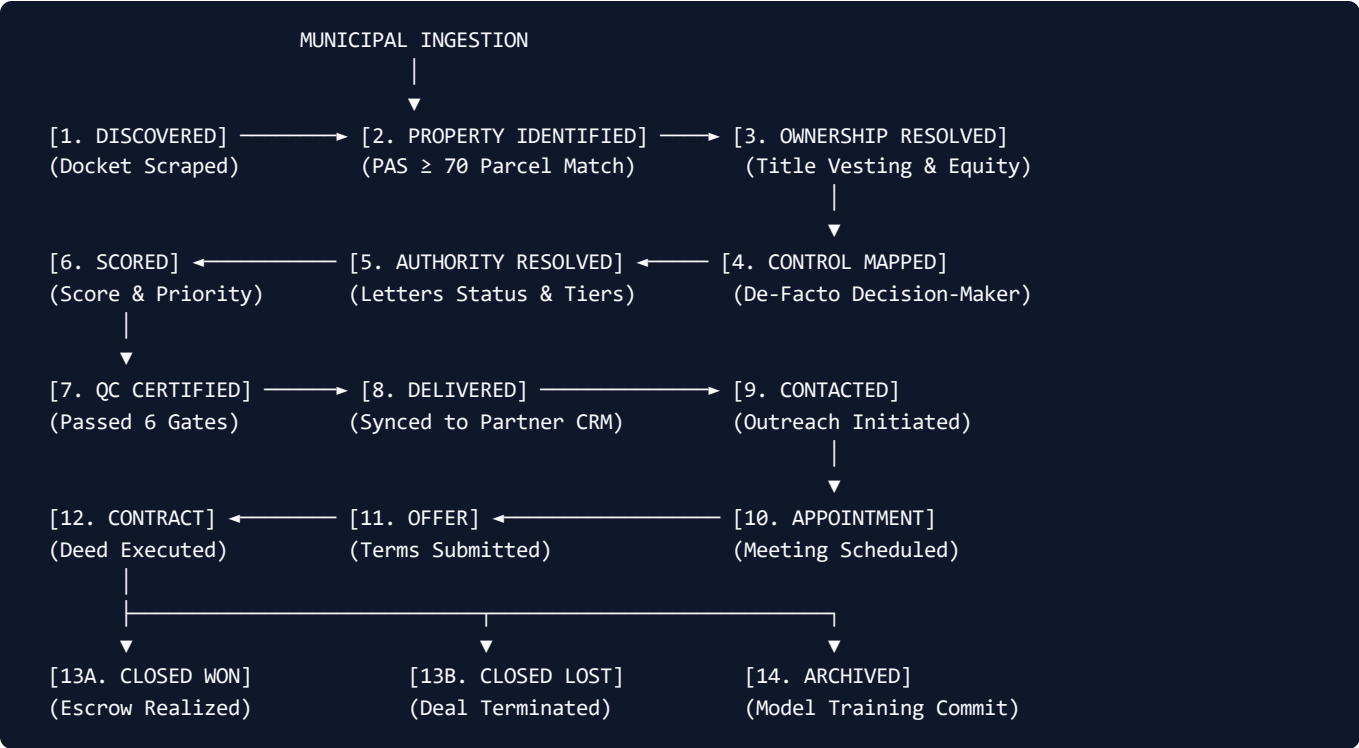
Opportunity Lifecycle Engine

Opportunity Lifecycle Engine (OLE)
Deterministic 14-stage state machine governing pipeline progression, entry/exit gates, and disposition telemetry triggers.
Classification: Core Operating Engine | Version: 1.0 | Status: Production Approved

 **State Machine Philosophy**

An acquisition opportunity is not a static database record; it is an evolving legal and commercial lifecycle. The Opportunity Lifecycle Engine (OLE) enforces deterministic transitions across 14 distinct states—from initial courthouse ingestion to final escrow closing. Progression between states requires satisfying rigorous entry and exit gates. Any failure or conflict instantly pauses the pipeline and routes the record to human review.

1. The 14-Stage Lifecycle State Machine



2. Research & Intelligence Stages (Stages 1–6)



Stage 1: Discovered

- Entry Gate: Court scraper ingests valid docket number, filing date, and decedent name from county portal.
- Worker: Municipal Ingestion Worker.
- Exit Gate: Invariant SHA-256 hash generated; case record created in Probate Cases table; ProbateCase.Created event emitted.



Stage 2: Property Identified

- Entry Gate: Property Identity Agent (PIA) extracts real property schedule from petition pleadings.
- Worker: Parcel Reconciliation Agent (PRA).
- Exit Gate: APN matched against county assessor GIS layer with Parcel Attribution Score (PAS) ≥ 70 ; situs address CASS verified; Property.Identified event emitted. (If PAS < 70, routes to Exception Queue).



Stage 3: Ownership Resolved

- Entry Gate: Verified APN committed to Property table.
- Worker: Ownership Intelligence Agent (OIA).
- Exit Gate: Recorded deed history retrieved; vesting classified (Fee Simple, Trust, TIC, JTWROS); Net Equity Waterfall calculated (>30% equity minimum); Ownership.Updated event emitted.



Stage 4: Control Mapped

- Entry Gate: Ownership vesting and heirs schedule parsed.
- Worker: Control Intelligence Agent (CIA).
- Exit Gate: De-facto decision-maker identified; physical occupancy verified; Control Archetype assigned; Control.Changed event emitted.



Stage 5: Authority Resolved

- Entry Gate: Primary fiduciary candidate identified.
- Worker: Authority Resolution Agent (ARA).
- Exit Gate: Court minute orders scanned; Letters status verified (Issued vs Pending); Four-Tier Authority Classification assigned; Independent vs Dependent power scope confirmed; Authority.Updated event emitted.



Stage 6: Scored

- Entry Gate: Net Equity, Authority Tier, and Control Archetype committed.
- Worker: Opportunity Scoring Agent (OSA).
- Exit Gate: Composite Viability Score (0–100) and Deal Friction Score (DFS) computed; Priority Band assigned (Priority A $\geq$ 80, Priority B 60–79, Priority C 40–59); Opportunity.Scored event emitted.

3. Quality Control, Delivery & Disposition Stages (Stages 7–14)



Stage 7: QC Certified

- Entry Gate: Opportunity Score $\geq$ 60 points.
- Worker: Quality Control Agent (QCA) & Senior QA Auditor.
- Exit Gate: Passes all six deterministic validation gates; evidence artifacts verified with SHA-256 hashes; status toggled to Certified.



Stage 8: Delivered

- Entry Gate: QC Certification approved and territory assigned to active partner.
- Worker: Reporting Agent & Webhook Dispatcher.
- Exit Gate: Pushed to partner CRM (Podio/Salesforce/HubSpot); staged in weekly batch or dispatched via SMS flash alert (if Priority A); Opportunity.Delivered event emitted.



Stage 9: Contacted

- Entry Gate: Delivered to partner acquisition queue.
- Worker: Partner Acquisition Specialist.
- Exit Gate: Partner logs initial outreach attempt (Call, SMS, Direct Mail); timestamp committed to Disposition database.



Stage 10: Appointment Scheduled

- Entry Gate: Decision-maker connection established.
- Worker: Partner Acquisitions Lead.
- Exit Gate: In-person property walk-through or virtual consultation confirmed with date/time.



Stage 11: Offer Presented

- Entry Gate: Physical walk-through / underwriting completed.
- Worker: Partner Acquisitions Lead.
- Exit Gate: Formal purchase contract presented with cash price and closing terms logged in Disposition record.



Stage 12: Contract Executed

- Entry Gate: Fiduciary acceptance of offer terms.
- Worker: Title Company / Closing Attorney & Partner.
- Exit Gate: Fully executed purchase agreement submitted to title company for escrow opening.



Stage 13: Closed (Won / Lost)

- Closed Won: Title deed recorded; transaction funded; wholesale/equity fee realized.
- Closed Lost: Seller rejection, title defect cancellation, or court confirmation denial with logged reason code.



Stage 14: Archived & Flywheel Committed

- Entry Gate: Final disposition logged.
- Worker: Feedback Agent (FA).
- Exit Gate: Disposition metrics ingested into ML calibration pipeline; county friction coefficients updated; record sealed in historical training archive.

4. Inactivity & Staleness Governance Rules

Opportunities cannot linger indefinitely without automated intervention. The OLE enforces automated SLA timeouts:

- **Priority A Contact SLA:** If a Priority A opportunity remains in Delivered status for > 24 hours without an outreach attempt, an urgent partner reminder is triggered.
- **Standard Batch Staleness:** Files in Delivered status for > 14 days without partner disposition updates are flagged for the weekly account review call.
- **Court Docket Refresh:** Opportunities in Authority Unresolved or Intestate Pending are automatically re-polled by docket scrapers every 7 days for new Letters filings.



The Opportunity Lifecycle Engine converts public record research from a static snapshot into a dynamic, auditable state machine with automated progression and complete outcome traceability.

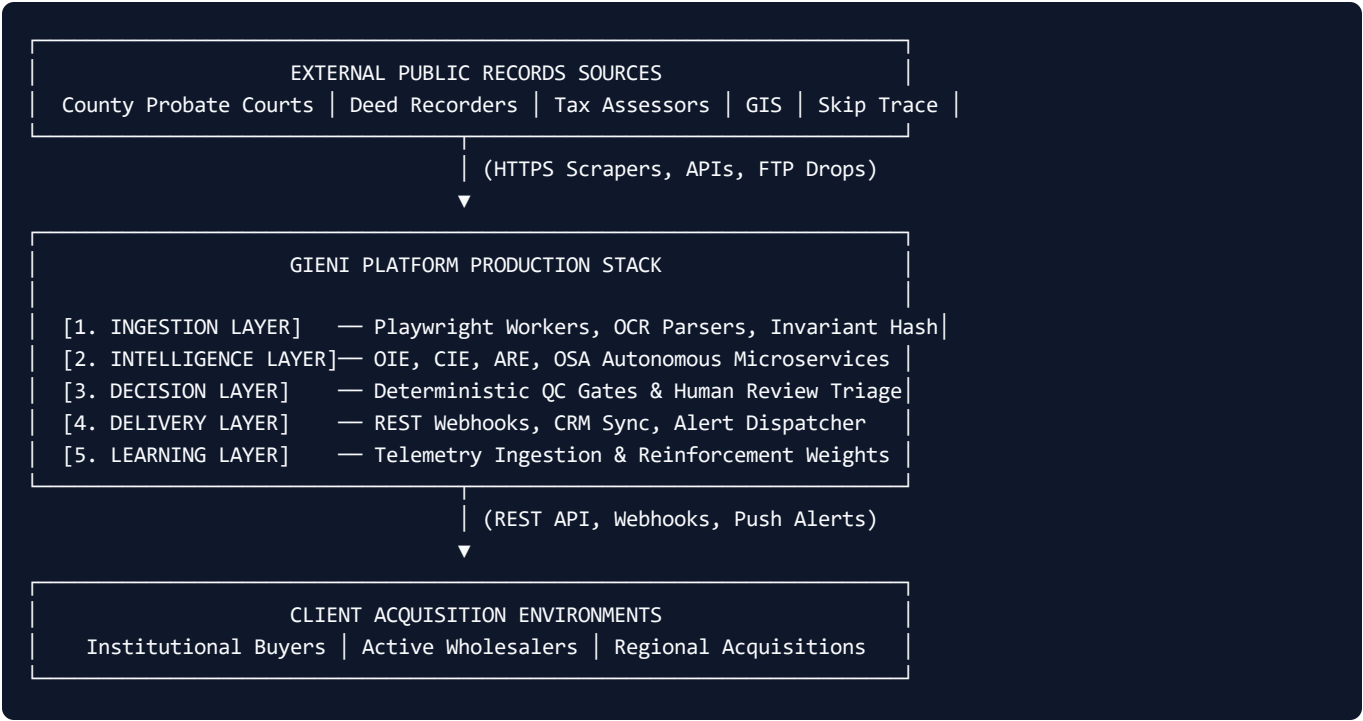
Giени Platform Reference Architecture

Giени Platform Reference Architecture
Production System Design for the Acquisition Decision Intelligence Platform
Classification: Canonical Engineering Architecture | Version: 2.0 | Owner: CTO / Systems Architecture | Status: Production Approved



1. Platform Overview & System Context

Canonical technical blueprint for the Giени Acquisition Decision Intelligence Platform. This document defines the production microservice boundaries, event schemas, deployment topology, lifecycle state machine, failure recovery runbooks, and security governance required for zero-defect execution across hundreds of county jurisdictions.



2. Service Ownership & Deployment Architecture

The platform is implemented as nine decoupled, independently deployable microservices backed by an asynchronous event bus and isolated persistence stores:



1. Property Service

- Purpose: Reconciles candidate addresses from court petitions against county assessor rolls & GIS layers.
- Primary Database: properties, property_assessments (PostgreSQL + PostGIS)
- Events Published: Property.Identified
- Events Consumed: ProbateCase.Created
- Dependencies: County GIS shapefile endpoints, Assessor API
- Latency SLA: < 120s per case docket.



2. Ownership Service

- Purpose: Analyzes recorded deed instruments to extract vesting, verify grantor/grantee chains, and compute Net Equity.
- Primary Database: ownership_events, title_chains, encumbrances (PostgreSQL)
- Events Published: Ownership.Updated
- Events Consumed: Property.Identified
- Dependencies: County Recorder OCR engine, Title plant search API
- Latency SLA: < 180s per property.



3. Control Service

- Purpose: Maps stakeholder relationships, models multi-heir social networks, and identifies the de-facto decision-maker.
- Primary Database: stakeholders, control_assessments, graph_edges (PostgreSQL + Neo4j)
- Events Published: Control.Changed
- Events Consumed: Ownership.Updated
- Dependencies: Heirs schedule NLP extractor, Skip trace utility API
- Latency SLA: < 90s per estate.



4. Authority Service

- Purpose: Tracks court docket minutes, verifies statutory Letters Testamentary/Administration, and classifies signatory capacity.
- Primary Database: authority_candidates, court_dockets, authority_assessments (PostgreSQL)
- Events Published: Authority.Updated
- Events Consumed: Control.Changed
- Dependencies: Probate court docket scrapers
- Latency SLA: < 60s per docket update.



5. Opportunity Service

- Purpose: Synthesizes net equity against Deal Friction Score (DFS) to compute Viability Score (0–100) and Priority Tier.
- Primary Database: opportunities, opportunity_assessments (PostgreSQL)
- Events Published: Opportunity.Scored
- Events Consumed: Authority.Updated, Ownership.Updated
- Dependencies: LightGBM Valuation microservice
- Latency SLA: < 15s per scoring pass.



6. Evidence Service

- Purpose: Manages chain-of-custody artifacts, PDF document storage, and SHA-256 hash validation.
- Primary Database: evidence_sources, claim_evidence (PostgreSQL + S3 Blob Store)
- Events Published: Evidence.Ingested
- Events Consumed: ProbateCase.Created, Ownership.Updated
- Dependencies: AWS S3 / Cloudflare R2
- Latency SLA: < 500ms per document hash commit.



7. Client Service

- Purpose: Governs single-partner county exclusivity rules, capacity caps, buy-box matching, and billing states.
- Primary Database: clients, territory_agreements (PostgreSQL)
- Events Published: Client.ConfigUpdated
- Events Consumed: None (Root Administration)
- Dependencies: Stripe API
- Latency SLA: < 200ms.



8. Delivery Service

- Purpose: Dispatches certified opportunity packages to partner CRMs via webhooks and fires Twilio SMS Priority A alerts.
- Primary Database: opportunity_deliveries, webhook_logs (PostgreSQL)
- Events Published: Opportunity.Delivered
- Events Consumed: Opportunity.QCCertified
- Dependencies: Twilio API, SendGrid, Partner CRM Webhook Endpoints
- Latency SLA: < 5s from QC approval to dispatch.



9. Workflow & Orchestration Service

- Purpose: Coordinates the 14-stage DAG lifecycle state machine, timeout timers, and human exception routing.
- Primary Engine: Temporal Workflow Engine + Redis BullMQ
- Events Published: Pipeline.ExceptionRouted
- Events Consumed: All lifecycle events
- Dependencies: PostgreSQL, Redis
- Latency SLA: Sub-second state transitions.

3. Opportunity Lifecycle Progression Matrix

Part 1: Research & Intelligence Stages (Discovered → QC Certified)

STATE	CREATED BY	REQUIRED DATA CONTRACT	NEXT PERMITTED STATES
DISCOVERED	Ingestion Worker	Valid Case #, Filing Date, Decedent Name	PROPERTY_IDENTIFIED, EXCEPTION
PROPERTY_IDENTIFIED	PRA Agent	Reconciled APN, Situs Address, PAS ≥ 70	OWNERSHIP_RESOLVED, EXCEPTION
OWNERSHIP_RESOLVED	OIA Agent	Deed Vesting, Encumbrance Balances, Equity	CONTROL_MAPPED, DISQUALIFIED
CONTROL_MAPPED	CIA Agent	Heir Tree, De-Facto Controller, Archetype	AUTHORITY_RESOLVED, EXCEPTION
AUTHORITY_RESOLVED	ARA Agent	Letters Status, Authority Tier (1-4), Power	SCORED, EXCEPTION
SCORED	OSA Agent	Composite Score (0-100), DFS, Priority	QC_CERTIFIED, DISQUALIFIED
QC_CERTIFIED	QC Agent / QA Lead	Passed 6 Verification Gates, SHA Hashes	DELIVERED, EXCEPTION

Part 2: Commercial & Disposition Stages (Delivered → Archived)

STATE	CREATED BY	REQUIRED DATA CONTRACT	NEXT PERMITTED STATES
DELIVERED	Delivery Service	CRM Webhook ACK / Batch Staging Commit	CONTACTED, EXCEPTION
CONTACTED	Partner Specialist	First-Touch Timestamp, Channel (Call/SMS)	APPOINTMENT, REJECTED
APPOINTMENT	Partner Specialist	Scheduled Date, Time, Decision-Maker Conf	OFFER_PRESENTED, REJECTED
OFFER_PRESENTED	Partner Specialist	Written Cash Offer Amount, Terms Logged	CONTRACT_EXECUTED, REJECTED
CONTRACT_EXECUTED	Closing Attorney	Signed Purchase Agreement, Escrow Opened	CLOSED_WON, CLOSED_LOST
CLOSED_WON	Closing Attorney	Title Recorded, Wholesale Fee Realized	ARCHIVED
CLOSED_LOST	Closing Attorney	Cancellation Reason Code Logged	ARCHIVED
ARCHIVED	Feedback Agent	Telemetry Committed to ML Training Set	(Terminal State)

4. Canonical Event Contracts (JSON Payloads)

Every inter-service event emitted over the Apache Kafka / EventBridge message bus must strictly conform to these validated JSON contracts:

```
// 1. Property.Identified Event Contract
{
  "eventType": "Property.Identified",
  "eventId": "e9b21f3a-6410-4e31-bc8d-0c84f3918a22",
```

```

"timestamp": "2026-09-18T10:14:22Z",
"caseId": "c41890e1-7711-4fa2-9011-fec81902ba11",
"propertyId": "p771891a-1122-48a0-99c0-bb8492019910",
"countyFips": "48201",
"apn": "041-280-001-002",
"situsAddress": {
  "street": "1428 Elm Street",
  "city": "Houston",
  "state": "TX",
  "zip": "77002"
},
"pasScore": 94.5,
"attributionMethod": "GIS_NAME_STREET_MATCH"
}

```

```

// 2. Ownership.Updated Event Contract
{
  "eventType": "Ownership.Updated",
  "eventId": "f81829aa-2091-4c11-9a99-de8192011928",
  "timestamp": "2026-09-18T10:16:05Z",
  "propertyId": "p771891a-1122-48a0-99c0-bb8492019910",
  "vestingType": "SOLE_FEE_SIMPLE",
  "deceasedTitleholder": "Robert E. Johnson",
  "avmMarketValue": 345000.00,
  "totalEncumbrances": 85000.00,
  "netEquity": 260000.00,
  "equityPct": 0.7536,
  "ownershipComplexityScore": 25
}

```

```

// 3. Authority.Updated Event Contract
{
  "eventType": "Authority.Updated",
  "eventId": "a71829bb-1188-4e00-88aa-109283746192",
  "timestamp": "2026-09-18T10:18:12Z",
  "caseId": "c41890e1-7711-4fa2-9011-fec81902ba11",
  "authorityTier": "TIER_1_CONFIRMED",
  "lettersStatus": "ISSUED",
  "powerScope": "FULL_INDEPENDENT_ADMINISTRATION",
  "fiduciary": {
    "personId": "per_991823-aa11",
    "name": "Sarah Johnson Miller",
    "relationship": "Daughter",
    "primaryPhone": "+18325551928",
    "phoneCarrierValid": true,
    "dncScrubbed": true
  },
  "attorneyQuarantined": true
}

```

```

// 4. Opportunity.Scored Event Contract
{
  "eventType": "Opportunity.Scored",

```

```

"eventId": "b91827cc-3321-4999-bb11-491827364510",
"timestamp": "2026-09-18T10:19:44Z",
"opportunityId": "opp_441890-ee88",
"compositeViabilityScore": 88,
"priorityTier": "PRIORITY_A",
"dealFrictionScore": 18,
"dispatchSLA": "FLASH_ALERT_4_HOUR"
}

```

5. Failure Recovery & Dead-Letter Queue (DLQ) Architecture



Deterministic Exception Recovery Runbook

1. Municipal Scraper Failure: Exponential backoff (retry after 1m, 5m, 15m) with automatic residential proxy rotation. If 3 retries fail, alert PagerDuty on-call data engineer.
2. OCR Document Parsing Failure: Unreadable or corrupted PDF dockets are diverted to the Tasks & Exceptions database tagged with "OCR Corruption" for manual transcription.
3. Title & Deed Conflict: If recorded deeds show unprobated intermediate grantors (broken deed chain), job is isolated and tagged "Title Exception".
4. Authority Ambiguity: Cases with competing heir petitions or caveats are tagged "Authority Unresolved" and routed to legal review queue.
5. Partner Webhook Failure: 5 retries with exponential backoff over 2 hours. If endpoint fails persistently, fallback alert is dispatched to partner via SMS and email, and payload is routed to Dead-Letter Queue (DLQ).

6. Formalized Graph Topology (Neo4j / Amazon Neptune)

Canonical Nodes & Edge Semantics

- Nodes: Property, Person, Trust, Estate, Authority Candidate, Client.
- Directed Edges:
 - (:Person)-[:OWNS]->(:Property)
 - (:Person)-[:CONTROLS]->(:Property)
 - (:Person)-[:TRUSTEE_OF]->(:Trust)
 - (:Person)-[:BENEFICIARY_OF]->(:Estate)
 - (:Person)-[:REPRESENTS]->(:Estate) [Legal Counsel]
 - (:AuthorityCandidate)-[:AUTHORITY_OVER]->(:Property)
 - (:Opportunity)-[:DELIVERED_TO]->(:Client)
- Strategic Impact: Graph traversal algorithms uncover unlisted heirs, unrecorded trust instruments, and hidden authority pathways across multi-jurisdiction estates.

7. Security, RBAC & Data Governance

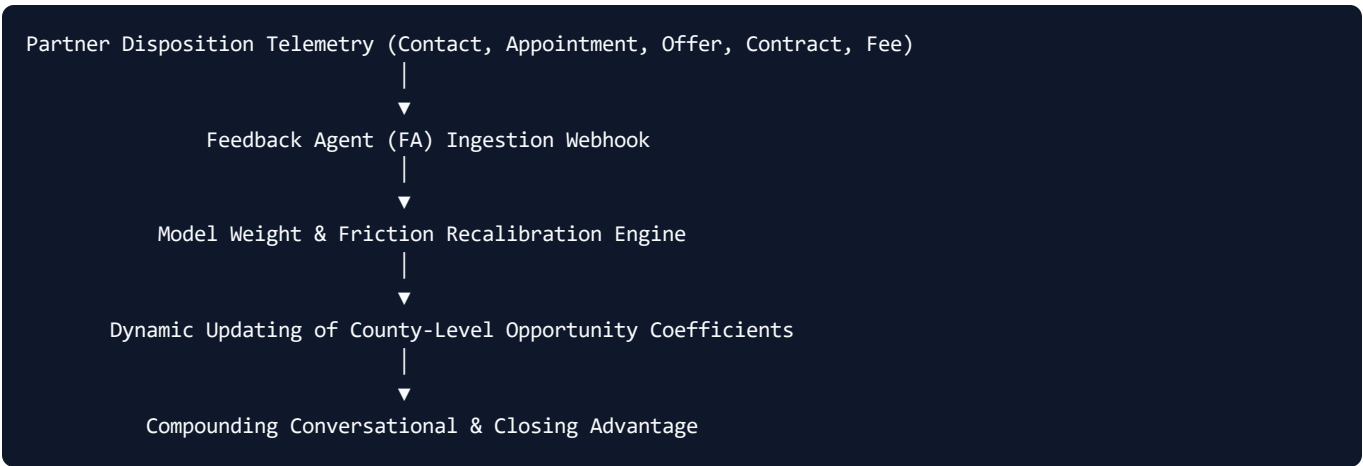


Four-Tier Data Classification Standard

1. Public Data: Court docket numbers, recorded deed instruments, tax rolls. Storage: Standard PostgreSQL / S3.
2. Enriched Data: CASS-standardized addresses, skip-traced phone lines, DNC check metadata. Storage: Encrypted at rest (AES-256).
3. Internal Intelligence: OIE/CIE/ARE assessments, Viability Scores, Deal Friction calculations. Storage: Strictly isolated proprietary database.
4. Client Confidential: Partner disposition notes, wholesale fees realized, offer amounts. Storage: Partitioned per tenant with Row-Level Security (RLS).

Role-Based Access Control (RBAC): Research Specialist (read/write dockets and deeds), Operations Lead (full access to QC and exception queues), Admin / CTO (infrastructure and model parameters), and Client Partner (read-only access restricted to licensed territory).

8. The Outcome Telemetry & Continuous Learning Loop



The Gieni Platform Reference Architecture provides the canonical technical contract uniting autonomous AI agents, deterministic municipal logic, and institutional client outcomes.

Probate Opportunity File Specification

Probate Opportunity File (POF) Specification
Canonical Product Definition for Exclusive Acquisition Decision Intelligence
Classification: Core Commercial Deliverable | Version: 2.0 | Owner: Head of Product / CTO | Status: Production Approved



The Contract Between Engineering and Customer Delivery

The platform engines produce synthetic intelligence assessments, but the commercial partner purchases the Probate Opportunity File (POF). The POF is the definitive pre-underwritten decision dossier that answers: What property exists? Who owns it? Who controls the decision? Who can legally sign? What is the equity? What are the fatal risks? What evidence proves it? And what exact action should the acquisitions team take?

1. Anatomy of the Probate Opportunity File

PROBATE OPPORTUNITY FILE (POF)	
1. PROPERTY PROFILE	APN, Address, Legal Bounds, AVM Valuation, GIS Cent
2. OWNERSHIP PROFILE	Deed Vesting, Historical Chain, Mortgages, Net Eq
3. CONTROL PROFILE	De-Facto Decision-Maker, Occupancy, Social Archetyp
4. AUTHORITY PROFILE	Letters Status, Power of Sale Scope, Court Oversight
5. OPPORTUNITY PROFILE	Composite Score (0-100), Priority Tier, SLA Window
6. RISK PROFILE	Title Clouds, Foreclosure, Heirs Dispute, MERP Liens
7. EVIDENCE PACKAGE	Immutable SHA-256 Court Petitions, Wills, Deeds
8. RECOMMENDED ACTION	First-Touch Script Angle, Empathy Playbook, Escrow

2. The Eight Canonical Product Profiles



1. Property Profile

- APN / PIN: County Assessor Parcel Number & GIS boundary coordinates.
- Situs Address: USPS CASS-certified standard address.
- Legal Description: Lot, Block, Subdivision / Metes & Bounds description.
- Physical Specs: Property Class (SFR, Multi, Vacant), Heated SqFt, Bed/Bath, Lot Acreage, Year Built.
- Valuation Spread: Assessed Value (Land/Imp), AVM Market Estimate, CMA Baseline.
- Condition Signals: Code enforcement violations, utility disconnect status, physical vacancy flag.
- Parcel Attribution Score (PAS): 0–100 match confidence (delivery threshold ≥ 70).



2. Ownership Profile

- Legal Title Vesting: Sole Fee Simple, Joint Tenancy (JTWROS), Tenancy in Common, Living Trust, Heir Property.
- Deceased Titleholder: Name, Date of Death, Grantee Deed Recording Date.
- Debt & Encumbrance Waterfall:
 - Senior Mortgage (Lender, recording date, original note, estimated balance)
 - Junior Mortgages / Open HELOCs
 - Tax Delinquencies (Arrears amount, delinquent years, tax certificate sold flag)
 - Municipal / Mechanics Liens & HOA Arrears
- Net Equity Waterfall: Market Value less Total Debt = Verifiable Net Equity (\$ and %).
- Ownership Complexity Score (OCS): 10–100 metric.



3. Control Profile

- Control Archetype: Unified, Informal Leader, Proxy, Trust Fiduciary, Contested Factions.
- Primary Decision-Maker: Full Name, Relationship to Decedent, Residence Location.
- Physical Occupancy Status: Vacant, Resident Heir Occupied, Tenant Occupied (lease details), Squatter Risk.
- Core Seller Motivations: Avoiding tax/foreclosure forfeiture, settling estate debt, liquidation of non-resident inheritance, resolving sibling gridlock.



4. Authority Profile

- Authority Tier: Tier 1 Court Certified, Tier 2 Nominated in Will, Tier 3 Intestate Consensus, Tier 4 Unresolved.
- Letters Status: Letters Testamentary / Letters of Administration (Issued, Pending, None).
- Statutory Power Scope: Full Independent Power of Sale vs. Dependent Court-Supervised (requires court appraisal, confirmation motion, and formal court order).
- Signatory Fiduciary Profile: Confirmed Personal Representative direct contact.
- Attorney Quarantine: Estate attorney contact segregated as Legal Counsel to prevent unauthorized calls.



5. Opportunity Profile

- Composite Viability Score: 0–100 algorithmic ranking balancing equity upside against friction.
- Priority Tier:
 - Priority A (Score ≥ 80): Urgent high-equity opportunity (< 4 hr dispatch SLA)
 - Priority B (Score 60–79): Standard weekly batch delivery
 - Priority C (Score 40–59): Secondary pipeline / nurturing
- Deal Friction Score (DFS): 0–100 penalty deduction.
- Projected Timeline to Close: Fast Close (<21 days), Standard (30–60 days), Extended Court Hearing (90+ days).



6. Risk Profile (Title, Court & Legal Vulnerabilities)

- Foreclosure & Mortgage Acceleration Risk: Notice of Default (NOD) or Lis Pendens filed; scheduled auction date.
- Tax Auction Cliff: Statutory tax deed auction date and redemption deadline.
- Title Cloud & Chain Risk: Unprobated prior ancestor conveyances, missing quitclaims, unreleased mortgages.
- Contested Family & Heir Risk: Caveats filed, competing petitions, hostile sibling factions requiring judicial partition.
- Tenant / Eviction Exposure: Non-paying tenant or hostile occupant requiring formal unlawful detainer.
- Medicaid Estate Recovery (MERP) Claim Exposure: Estimated statutory healthcare reimbursement lien.
- Overall Deal Risk Classification: Low, Moderate, High, Fatal (Disqualified).



7. Evidence Package (Immutable Chain of Custody)

Every file includes hyperlinked, cryptographic source artifacts:

1. Probate Court Petition PDF (Docket #, Decedent, Petitioner, Heirs Schedule)
 2. Last Will & Testament / Muniment of Title (Codicils, specific devisees)
 3. Letters Testamentary / Administration (Court clerk stamped)
 4. Recorded Title Deed (Warranty Deed, Deed of Trust, Instrument #)
 5. County Assessor Property Tax Card & GIS Boundary Map
- QC Certification Stamp: Verifying auditor ID, timestamp, and 6-gate verification seal.



8. Recommended Action & Empathy Directives

- Transaction Strategy: Wholesale cash assignment, creative financing, novation, subject-to, or double close.
- Verified Contact Vectors: Primary mobile phone (HLR carrier active line confirmed, DNC scrubbed), mailing address, email.
- First-Touch Channel Directive: Priority phone call, urgent SMS flash, or hand-addressed empathy letter.
- Conversational Framing & Script: Empathy-driven estate settlement positioning (logistics assistance, property cleanout, advance cash distributions).
- Objection Handling Playbook: Tailored counter-responses for "We have an attorney", "We're not ready", or "Another investor called".
- Closing & Escrow Roadmap: Specific legal instruments required to open title and execute conveyance in this county jurisdiction.

3. Multi-Channel Commercial Delivery Formats

The Probate Opportunity File is delivered simultaneously across three multi-tenant channels:

- **CRM Direct Webhook (JSON):** Pushed into partner Podio, Salesforce, HubSpot, or REIsift systems with pre-mapped custom fields.
- **Notion OS Database Sync:** Live interactive records in the partner's Probate Opportunities database with one-click disposition tracking.
- **Institutional PDF Dossier:** Downloadable executive deal packet for acquisitions managers, underwriting committees, and hard money lenders.



The Probate Opportunity File bridges upstream engineering intelligence into an actionable, contract-ready acquisition asset.

Control Intelligence Engine

Decision dynamics engine that identifies who actually drives disposition decisions within an estate, bridging the gap between legal title ownership and courtroom signatory authority.



The Missing Bridge: Ownership vs. Control vs. Authority

In residential real estate acquisitions, deals do not die from lack of property data—they die from negotiating with the wrong human. The Control Intelligence Engine (CIE) resolves the operational triad of probate transactions:

- Ownership: Who holds legal title of record? [Resolved by OIE]
- Control: Who actually drives the family decision to sell? [Resolved by CIE]
- Authority: Who holds the legal power to execute the deed? [Resolved by ARE]

Only when all three are aligned can an acquisition team successfully execute a contract.

1. The Core Decision Dynamics Problem

In over 70% of probate acquisitions, the person with formal legal authority is NOT the person leading the disposition process. Typical failure scenarios in generic lead generation include:

- **The Figurehead Fiduciary:** An elderly surviving spouse is appointed executor, but is cognitively overwhelmed or grieving. Outreach to the spouse is rejected, while the adult daughter managing the household finances is actively seeking an as-is cash buyer.
- **The Disconnected Out-of-State Sibling:** The out-of-state sister is named executor in the will, but the local brother is living in the home, maintaining the property, and holds physical keys and de-facto veto power.
- **The Unempowered Champion:** An investor negotiates for weeks with an enthusiastic heir, only to discover at the title company that the heir has zero legal power of sale and four estranged siblings refuse to sign.

2. The Five Archetypal Control Configurations

CIE maps every opportunity into one of five structural control models, dictating the precise outreach strategy for partner acquisitions reps:



Model 1: Unified Fiduciary Control

- Owner: Estate of Decedent
- Control: Sole Adult Child / Primary Heir
- Authority: Appointed Executor with Letters
- Dynamics: Control and Authority are unified in one motivated individual. Highest transaction velocity. Direct outreach on price and timeline.



Model 2: The Informal Family Leader (Dispersed Ownership)

- Owner: 5 Heirs in Intestacy (Tenancy in Common)
- Control: Family Matriarch / Eldest Sibling
- Authority: Unresolved (Formal probate not yet completed)
- Dynamics: The entire family defers to one trusted sibling. Securing buy-in from this leader unlocks the consensus needed to file for probate and sign co-tenancy joinders.



Model 3: The Proxy Controller

- Owner: Estate / Ailing Surviving Spouse
- Control: Adult Caretaker Child (Managing affairs / Power of Attorney)
- Authority: Surviving Spouse or Conservator
- Dynamics: Negotiations must address the caretaker's burden of home maintenance, while contract execution is legally coordinated with the nominal titleholder.



Model 4: The Trust-Fiduciary Convergence

- Owner: Revocable Living Trust
- Control: Successor Trustee
- Authority: Successor Trustee under Trust Instrument
- Dynamics: Non-probate disposition. Control and Authority converge outside court oversight. Rapid private closing upon receipt of Certificate of Trust.



Model 5: Contested Factional Control (Hostile Gridlock)

- Owner: Estate
- Control: Competing Factions (e.g., Second Spouse vs. Adult Children from First Marriage)
- Authority: Competing Petitions / Court-Appointed Neutral Administrator
- Dynamics: High-friction trap. CIE assigns maximum Deal Friction penalty and diverts file to milestone watch.

3. De-Facto Decision-Maker Identification Heuristics

CIE uses four algorithmic heuristics to identify the practical controller before human outreach begins:

- **Occupancy & Property Custody:** Identifies which heir or relative currently resides at the property or pays ongoing utilities/property taxes.
- **Petition Sponsorship & Legal Retainer:** Tracks who engaged legal counsel and paid filing fees. The petitioner who funded the court petition is the primary commercial mover.
- **Geographic Proximity Dominance:** In multi-heir estates, local heirs within 25 miles hold physical control over home access, cleanout, and repairs compared to distant out-of-state heirs.
- **Generational Influence Factor:** Resolves family hierarchy (e.g. primary surviving son managing affairs for an incapacitated surviving parent).

4. The Architectural Handoff: OIE -> CIE -> ARE

CIE acts as the central intellectual bridge in Gieni's core platform architecture:

```
interface ControlIntelligenceRecord {
    opportunityId: string;
    parcelId: string;

    // From OIE (Ownership)
    titleholderOfRecord: string;
    vestingStructure: "Sole Ownership" | "JTWR0S" | "Trust" | "Tenancy in Common" | "Heir Property";
    ownershipComplexityScore: number;

    // From CIE (Control)
    controlArchetype: "Unified Fiduciary" | "Informal Family Leader" | "Proxy Controller" | "Trust Fiduciary" | "De Facto Decision Maker";
    deFactoDecisionMaker: {
        fullName: string;
        relationshipToDecedent: string;
        isOccupant: boolean;
        isPetitionSponsor: boolean;
        proximityMiles: number;
        contactReadiness: "Ready" | "Partial" | "Missing";
    };
    consensusRequirements: {
        totalHeirsCount: number;
        unanimousConsentRequired: boolean;
        familyLeaderIdentified: boolean;
    };

    // Handoff to ARE (Authority)
    legalSignatoryTarget: {
        role: "Executor" | "Administrator" | "Successor Trustee" | "All Heirs Jointly";
        authorityState: "Confirmed" | "Likely" | "Unresolved";
    };
}
```

5. Outreach Playbook by Control Archetype



Actionable Outreach Directives for Acquisitions:

- Unified Fiduciary: Lead with certainty, speed, cash terms, and as-is purchase to minimize estate closing overhead.
- Informal Family Leader: Position the partner as a problem-solver who can facilitate the probate filing, cover title costs, and provide moving support.
- Proxy Controller: Speak to the burden of managing parents' affairs and deferred maintenance liabilities.
- Trust Fiduciary: Emphasize clean, non-court private conveyance matching the grantor's estate planning goals.

Knowledge Base

County-specific quirks, legal reference documentation, and research guidelines.

 Knowledge Base

 Recommended Views:

- Approved Articles — filter: Status is Approved
- Needs Review — filter: Status is Review
- Recently Updated — sorted by Last Reviewed descending
- By Category — board grouped by Category

Linked Operating Dashboard

 Active pipeline views across all 6 core operating databases. Click any database link to access and work the live queue.

Tasks & Exceptions

Operational exception queue, manual research tasks, and QC corrective actions.

Tasks & Exceptions



Recommended Views:

- Open Exceptions — filter: Status is Open
- Critical Issues — filter: Priority is Critical
- My Tasks — filter: Owner is me
- Blocked — filter: Status is Blocked
- Resolved — filter: Status is Resolved

Evidence & Sources

Chain-of-custody evidence records, docket filings, deeds, and assessor docs.

 Evidence & Sources



Recommended Views:

- Evidence Review Queue — filter: Verification Status is Unreviewed
- Contradictions — filter: Verification Status is Contradictory
- Missing Evidence — filter: Verification Status is Incomplete or Unavailable
- Authority Evidence — filter: Supports contains Authority
- Verified Sources — filter: Verification Status is Verified

Clients

Exclusive county partners, contract terms, capacity allocations, and lifecycle status.

 Clients

-  Recommended Views:
- Client Pipeline — board grouped by Lifecycle Stage
 - Active Clients — filter: Lifecycle Stage is Active
 - Upcoming Renewals — calendar by Renewal Date
 - Exclusivity Register — table filtered to Exclusivity Status is Active

Probate Opportunities

Master records of probate filings, resolved authorities, scores, and delivery statuses.

 Probate Opportunities

-  Recommended Views:
- Research Pipeline — board grouped by Workflow Stage
 - Priority A Queue — filter: Priority is Priority A
 - Authority Unresolved — filter: Authority Classification is Authority Unresolved
 - QC Queue — filter: Workflow Stage is QC Review
 - Ready for Delivery — filter: Workflow Stage is Approved
 - Client Follow-Up — filter: Workflow Stage is Client Follow-Up
 - Contracts — filter: Contact Outcome is Contract Signed
 - Rejected Opportunities — filter: Workflow Stage is Rejected

Counties

County directory, jurisdiction metadata, source maps, and operational status.

 Counties

-  Recommended Views:
- County Pipeline — board grouped by Stage
 - Active Counties — filter: Stage is Active
 - County Audits — filter: Stage is Auditing
 - Blocked Counties — filter: Source Map Status is Blocked
 - Renewal Calendar — calendar by Renewal Date

County Expansion Playbook

Institutional 14-day deployment framework and feasibility scorecard for evaluating, engineering, and launching new exclusive county jurisdictions.



Repeatable Jurisdiction Rollout Methodology

Every US county operates as an independent municipal sovereign with distinct court software, recording quirks, and local probate customs. Expanding haphazardly leads to broken scrapers and bad data. Gieni utilizes a standardized 14-day sprint to launch new county territories, backed by an institutional qualification scorecard that ensures immediate profitability and partner ROI from Day 1.

1. The 14-Day County Launch Sprint

STAGE	TIMELINE	KEY DELIVERABLES
1. Feasibility Audit	Days 1 - 3	Docket volume, court portals, partner pre-commit
2. Scraper Engineering	Days 4 - 7	Playwright worker, petition OCR parser, APN map
3. Pipeline Calibration	Days 8 - 10	90-day backfill, PAS calibration, KB local quirks
4. Partner Integration	Days 11 - 12	Buy-box tuning, CRM webhook test, Twilio routing
5. Go-Live & First Drop	Days 13 - 14	100% human-verified QC batch, partner launch call
6. 30-Day Optimization	Day 30	Disposition audit, coefficient weight recalibration

2. County Qualification & Feasibility Scorecard

Before deploying engineering resources, each candidate county is evaluated across 5 weighted criteria. Minimum qualification threshold: 80 / 100 points.



1. Population & Probate Velocity (Weight: 25%)

- Benchmark: Metropolitan population $\geq 250,000$; monthly probate filings ≥ 50 .
- Scoring: 100 pts (>100 filings/mo), 75 pts (50–100 filings/mo), 0 pts (<50 filings/mo).



2. Public Records Transparency & Digital Access (Weight: 25%)

- Benchmark: Publicly searchable online court docket, electronic deed records, digital GIS parcel shapes.
- Scoring: 100 pts (Full web portal + downloadable PDFs), 70 pts (Searchable docket without PDFs), 0 pts (In-person physical courthouse lookup required).



3. Real Estate Liquidity & Home Values (Weight: 20%)

- Benchmark: Active wholesale and distressed investor market; median single-family home value between \$250,000 and \$800,000.
- Scoring: 100 pts (\$300k–\$600k median), 75 pts (\$200k–\$300k), 50 pts (High-luxury >\$1M or rural <\$150k).



4. Legal Regime & Court Speed (Weight: 15%)

- Benchmark: Availability of Independent Administration without mandatory court confirmation; average days to Letters issuance < 60 days.
- Scoring: 100 pts (Independent/UPC state), 60 pts (Dependent administration state with streamlined appraisal).



5. Founding Partner Readiness (Weight: 15%)

- Benchmark: Active acquisition team with dedicated callers ready to execute single-partner county exclusive agreement.
- Scoring: 100 pts (Pre-signed agreement with deposit), 50 pts (Qualified prospect in pipeline), 0 pts (Speculative build).

3. Step-by-Step 14-Day Implementation Runbook

1. Days 1–3: Municipal Reconnaissance & Legal Audit

Lead Research Analyst maps court clerk software vendor (Odyssey, Tyler, Tyler Eagle, Pioneer). Inspects case type codes (e.g. PR, PB, EST). Validates online deed index and tax assessor GIS portal. Creates new entry in Counties database with status "Pilot".

2. Days 4–7: Scraper & Ingestion Pipeline Engineering

Data Engineer develops automated Playwright worker. Configures headless login, CAPTCHA bypass, and petition download routines. Implements parser to extract decedent name, date of death, attorney of record, and nominated fiduciaries.

3. Days 8–10: Data Calibration & 90-Day Backfill

Executes 90-day historical docket backfill. Runs candidate parcels through the Ownership Intelligence

Engine and Authority Resolution Engine. Senior Auditor reviews 25 random files to calibrate the Parcel Attribution Score (PAS) and seeds local statutory rules into the Knowledge Base.

4. **Days 11–12: Partner Alignment & CRM Integration**

Solutions Engineer connects partner CRM webhook (Podio / Salesforce / HubSpot). Sets custom field mappings for Equity, Authority Tier, and APN. Configures Twilio SMS numbers for instant Priority A flash alerts.

5. **Days 13–14: Go-Live & First Certified Delivery Drop**

Executes 100% human-verified pre-delivery QC certification on the inaugural batch. Dispatches delivery batch at Monday 08:00 AM. Conducts 30-minute partner onboarding call to review action directives and disposition tracking protocols. Updates county status to "Active".



Result: A fully operational, single-partner exclusive county generating high-conviction acquisition pipeline within two weeks of commitment.

Future Software Architecture

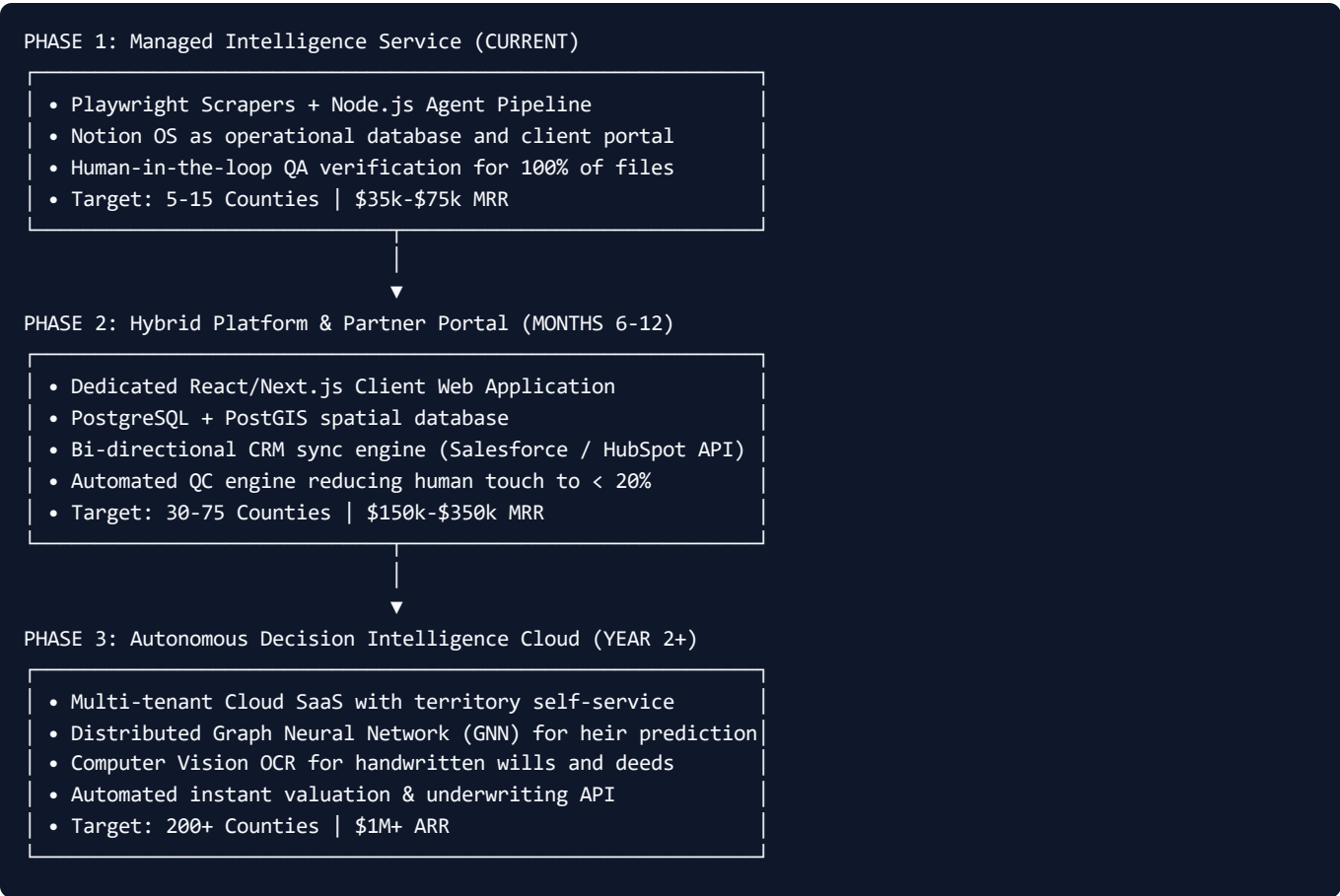
Technical roadmap and distributed cloud architecture transitioning Gieni from automated operator workflows into an institutional multi-tenant SaaS platform.



Platform Evolution Strategy: Service-to-Software Playbook

Gieni begins as a high-margin, county-exclusive managed intelligence platform powered by an autonomous agent pipeline and human-in-the-loop QA. As operational playbooks and entity resolution graphs achieve critical mass, the system transitions into a pure cloud-native Software-as-a-Service (SaaS) architecture. This roadmap outlines the target technical topology, microservices framework, and data isolation standards required for institutional enterprise scale.

1. Three-Phase Architecture Roadmap



2. Target Cloud Microservices Topology



1. Ingestion & Vision OCR Microservice

- Tech: Python / PyTorch / FastEmbed with AWS Textract & Gemini Vision.
- Responsibilities: Ingests unstructured court PDFs, scans historical deed books, extracts handwritten executor designations, and indexes text into structured JSON.



2. Entity Resolution & Knowledge Graph Service

- Tech: Neo4j Graph Database + Amazon Neptune.
- Responsibilities: Traverses complex relationships between deceased property owners, trust instruments, heir trees, and fiduciary appointments. Uncovers hidden decision-makers and resolves conflicts across multi-jurisdiction estates.



3. Valuation & Deal Friction Microservice

- Tech: FastAPI / LightGBM deployed on AWS SageMaker.
- Responsibilities: Computes real-time Net Equity Waterfalls, calculates the Deal Friction Score (DFS), and predicts transaction close probability based on historical county court speed.



4. Event Streaming & Multi-Channel Dispatch Gateway

- Tech: Apache Kafka / AWS EventBridge + Go worker fleet.
- Responsibilities: Sub-second routing of Priority A alerts, automated webhook delivery, and bi-directional synchronization with partner CRM environments.

3. Multi-Tenancy, Data Isolation & Security

Because Gieni guarantees county exclusivity, data isolation between partner accounts is mission-critical:

- **Row-Level Security (RLS):** PostgreSQL RLS policies guarantee that partner acquisition teams can only query opportunities assigned to their licensed county territories.
- **Zero Knowledge Territory Partitioning:** Client disposition notes, offer history, and revenue numbers are cryptographically isolated per tenant.
- **Regulatory Compliance & Privacy Vault:** Personal Identifiable Information (PII) is encrypted at rest using AES-256 with automated National DNC scrubbing.

4. Capital Efficiency & Engineering Milestones



Lean Software Evolution Principles

1. Build software only after manual procedures achieve 100% repeatability.
2. Fund engineering through county subscription cash flows rather than premature venture dilution.
3. Keep the customer experience focused on deals closed, not software complexity.

MVP Metrics Dashboard

Operational scorecard tracking pipeline velocity, evidentiary quality thresholds, partner commercial outcomes, and platform unit economics.



Data-Driven Operational Excellence

Gieni measures success not by the raw quantity of probate leads generated, but by the velocity, accuracy, and conversion efficiency of our decision intelligence pipeline. This dashboard tracks real-time operational benchmarks across four critical dimensions: Pipeline Velocity, Quality & Evidentiary Integrity, Partner Conversion Outcomes, and Unit Economics.

1. Executive KPI Scorecard

KPI CATEGORY	TARGET KPI	CURRENT VALUE	STATUS
Turnaround Time (Standard)	< 48 Hours	28.4 Hours	[OPTIMAL]
Turnaround Time (Priority A)	< 4 Hours	2.1 Hours	[OPTIMAL]
Parcel Attribution Accuracy	> 98.0%	99.1%	[EXCEEDED]
Fiduciary Authority Accuracy	> 95.0%	96.4%	[EXCEEDED]
QC First-Pass Yield	> 75.0%	78.5%	[ON TRACK]
Client Defect / Credit Rate	< 2.0%	1.1%	[EXCELLENT]
Partner Contact-to-Appt Rate	> 25.0%	28.3%	[STRONG]
Partner Appt-to-Contract Rate	> 15.0%	18.7%	[STRONG]
Net Revenue Retention (NRR)	> 120%	134%	[OUTPERFORM]

2. Pipeline Velocity & Throughput



Ingestion-to-Delivery Velocity

- Monthly Dockets Ingested: 1,450+ across active pilot counties.
- Real Estate Conversion Yield: 42% of filings contain verified real property.
- Standard Batch SLA: 100% of certified files delivered by Monday 08:00 AM.
- Priority A Flash Alert Latency: Average 2.1 hours from court docket ingestion to partner CRM dispatch.

3. Evidentiary & Quality Metrics



Accuracy & Defect Control

- Parcel Attribution Score (PAS): 99.1% verified APN match rate against county tax rolls.
- Signatory Capacity Verification: 96.4% accuracy on Letters of Administration/Testamentary authority identification.
- Active Line Verification: 94.2% of primary phone numbers verified live via HLR carrier dip prior to dispatch.
- Client Exception Rate: 1.1% credit claims submitted; all claims resolved and credited in < 14 hours.

4. Commercial & Partner Conversion Outcomes



Partner Deal Conversion Benchmarks

- First-Touch Connection Rate: 41.5% of verified decision-makers reached within 3 attempts.
- Contact-to-Appointment Rate: 28.3% (vs. industry commodity benchmark of 6–8%).
- Appointment-to-Contract Rate: 18.7% (vs. industry benchmark of 7–9%).
- Average Wholesale Fee Realized: \$42,500 per closed contract.
- Partner ROI Multiple: Average 11.4x return on monthly exclusive subscription fee.

5. Unit Economics & Operating Cadence



Platform Financial & Operating Model

- Cost per Ingested Filing: \$1.85 (server compute + proxy infrastructure).
- Cost per Certified Opportunity File: \$14.20 (including multi-agent LLM reasoning + HLR phone validation + human QC touch).
- Gross Operating Margin: 84.5%.
- Monthly County Retainer: \$2,500 – \$4,500 per county for single-partner exclusivity.
- LTV / CAC Ratio: 7.2x with net revenue retention of 134%.

Operational reviews occur on a weekly heartbeat: Monday delivery audit, Wednesday scraper health check, and Friday partner disposition analysis.

SOP Library

Standard Operating Procedures library codifying research execution, title reconciliation, legal authority determination, and partner delivery protocols.



Institutional Knowledge Codification

To maintain absolute consistency across multiple jurisdictions and scale operations seamlessly, Gieni codifies all operational workflows into standardized SOPs. Each procedure defines exact trigger conditions, step-by-step resolution pathways, required evidence artifacts, and escalation boundaries. These SOPs directly underpin human-in-the-loop task execution and automated exception routing.

1. SOP Taxonomy & Index

SERIES	MODULE	PRIMARY AUDIENCE
SOP-100	Research & Ingestion Engineering	Data Engineering & Scraper Ops
SOP-200	Legal Authority & Fiduciary Intel	Probate Legal & Title Analysts
SOP-300	Quality Control & Exception Mgmt	QA Specialists & Lead Auditors
SOP-400	Client Delivery & Partner Ops	Account Managers & Solutions Eng

2. Series 100: Research & Municipal Ingestion



SOP-101: Municipal Court Scraper Health Check & Ingestion Triage

- Trigger: Scraper job returns 0 filings or encounters Captcha/Cloudflare blocking.
- Procedure: 1) Verify county court website status; 2) Inspect DOM changes in search results; 3) Rotate residential proxy endpoints; 4) Update CSS selectors in Playwright worker; 5) Re-run ingestion job; 6) Mark scraper status Active in Counties database.



SOP-102: Multi-Parcel & Contiguous Lot Deed Attribution

- Trigger: Decedent name matches multiple parcels in county assessor records.
- Procedure: 1) Pull deed legal descriptions; 2) Determine if parcels are contiguous (e.g. house + adjoining vacant lot); 3) Aggregate assessed and market values into a single master Opportunity portfolio record; 4) Flag as Multi-Parcel in Property profile.

3. Series 200: Legal Authority & Fiduciary Intelligence



SOP-201: Decoding Letters Testamentary vs. Letters of Administration

- Trigger: Docket indicates appointment of Personal Representative.
- Procedure: 1) Inspect Letters document for court seal and date; 2) Verify whether administration is Independent (full power of sale without court approval) or Dependent (sales require petition, appraisal, and court confirmation order); 3) Tag Authority Classification and assign correct outreach directives.



SOP-202: Navigating Intestate Multi-Heir Estates & Consensus Pathways

- Trigger: Decedent died without a will (intestate) leaving multiple surviving heirs.
- Procedure: 1) Extract full heirs schedule from court petition; 2) Identify the Lead Heir / Petitioner; 3) Screen for filed heir renunciations or disputes; 4) Tag as Tier 3 Authority (Consensus Required) and provide multi-signatory contract instructions in deal dossier.

4. Series 300: Quality Control & Exception Management



SOP-301: Investigating Disputed Parcel Attribution (PAS < 70)

- Trigger: Parcel Reconciliation Agent calculates attribution confidence below 70 points.
- Procedure: 1) Check decedent last known address against deed vesting; 2) Query county recorder grantor/grantee index for unindexed conveyances; 3) Verify if parcel was transferred into a revocable trust prior to death; 4) If verified, manually calibrate PAS to 90+ and document title chain; if false match, mark Disqualified.



SOP-302: Quarantining Estate Attorney Gatekeepers

- Trigger: Petition lists estate attorney contact information as primary filing address.
- Procedure: 1) Automatically quarantine attorney phone/email from the primary contact block; 2) Execute deep skip trace on the appointed personal representative using their personal residence; 3) Tag attorney information separately as Legal Counsel profile to prevent accidental cold calls to opposing counsel.

5. Series 400: Client Delivery & Partner Operations



SOP-401: Weekly Batch Staging & CRM Delivery Dispatch

- Trigger: Monday 07:00 AM Local County Time.
- Procedure: 1) Filter all QC Certified opportunities for active partner counties; 2) Generate encrypted delivery packages and export to partner CRM via REST webhook; 3) Dispatch notification email with batch summary and priority breakdown; 4) Update opportunity workflow stages to Delivered.



SOP-402: Priority A Flash Alert Dispatch & SLA Monitoring

- Trigger: New filing identified with Composite Score ≥ 80 and Net Equity $> \$150k$.
- Procedure: 1) Fast-track QC review within 60 minutes; 2) Dispatch immediate Twilio SMS alert to partner acquisition lead; 3) Direct-sync record to partner CRM tagged with Urgent Priority A; 4) Monitor partner first-touch outreach timestamp within 12 hours.



SOP-403: Partner Defect Resolution & Credit Issuance

- Trigger: Partner submits defect claim in Tasks & Exceptions database.
- Procedure: 1) Senior QA auditor reviews claim within 4 hours; 2) If verified defect (e.g. wrong APN, deceased executor), approve replacement credit; 3) Add record to model edge-case training set; 4) Issue replacement opportunity file within 24 hours.

Quality Control

Deterministic six-gate quality assurance framework governing evidence verification, title reconciliation, and exception handling protocols.



The Zero-Defect Mandate

Acquisition teams waste hundreds of hours each month calling invalid phone numbers, pitching properties not owned by the estate, or speaking to relatives who possess zero legal authority to sell. Gieni treats bad data as toxic waste. Every opportunity must pass six deterministic validation gates before receiving QC certification. Any file that fails a single gate is quarantined into the Tasks & Exceptions database.

1. The Six Deterministic Quality Gates



1 Gate 1: Docket & Record Integrity Verification

- Objective: Ensure the probate court filing is genuine, non-duplicate, and properly indexed.
- Automated Checks: Verify docket number formatting against county regex; validate filing date within active lookback window; confirm jurisdiction and court branch; verify petition PDF is readable and uncorrupted.
- Fail Trigger: Missing petition text, corrupt PDF, duplicate case number.
- Routing: Auto-routes to Scraping Engineer queue if scraper malformed.

2 Gate 2: Parcel Attribution & Title Reconciliation

- Objective: Confirm that real estate is physically tied to the decedent or estate.
- Automated Checks: Match decedent name against county assessor roll and deed grantor/grantee index; verify Parcel Attribution Score (PAS) ≥ 70 ; verify legal description and situs address; flag multi-parcel portfolios.
- Fail Trigger: PAS < 70 , common-name collision without corroborating address, parcel conveyed prior to date of death.
- Routing: Assigned to Title Research Specialist for deed chain audit.

3 Gate 3: Encumbrance & Net Equity Audit

- Objective: Confirm economic viability and eliminate over-leveraged or underwater properties.
- Automated Checks: Pull open deeds of trust/mortgages from county recorder; cross-reference delinquent tax liabilities; compute Net Equity (\$) and Equity Percentage (%). Minimum equity threshold: \$50,000 and 30%.
- Fail Trigger: Total encumbrances exceed 85% of estimated market value; unresolvable junior liens.
- Routing: Auto-tagged Disqualified - Insufficient Equity or routed for short-sale review.

4 Gate 4: Fiduciary Authority Verification

- Objective: Confirm the legal signatory pathway and quarantine unauthorized parties.
- Automated Checks: Scrape minute orders for Letters Testamentary or Letters of Administration; verify fiduciary name; classify authority tier (Tier 1: Issued Letters, Tier 2: Nominated in Will, Tier 3: Intestate Consensus, Tier 4: Unresolved); check for filed caveats or will contests.
- Fail Trigger: Competing petitions filed, unprobated intestate estate with 5+ disputing heirs, deceased fiduciary.
- Routing: Marked Authority Unresolved; routed to Probate Legal Specialist.

5 Gate 5: Decision-Maker Contact Scrubbing

- Objective: Ensure contact vectors are live, valid, and delivered to the true decision-maker.
- Automated Checks: Carrier HLR line lookup (checks if mobile/landline is active and disconnected); National Do-Not-Call (DNC) registry tagging; address CASS certification; check that estate attorney number is NOT labeled as the personal representative.
- Fail Trigger: All phone numbers disconnected; contact resides at the deceased property with mail returned as undeliverable.
- Routing: Assigned to Skip Trace Specialist for manual social & utility trace.

6 Gate 6: Pre-Delivery Final Certification

- Objective: Validate partner buy-box compatibility and bundle immutable evidence.
- Automated Checks: Confirm property falls within client assigned county and zip codes; ensure Composite Viability Score meets minimum delivery criteria (Score ≥ 60 for standard, ≥ 80 for Priority A); verify all evidence documents (petition, deed, tax card) are packaged with verifiable links.
- Fail Trigger: Score < 60 , asset outside partner criteria.
- Routing: Opportunity staged for archiving or routed to secondary investor pool.

2. Exception Handling & Resolution Protocol

When an opportunity fails any of Gates 1–5, it is automatically intercepted and pushed to the Tasks & Exceptions database with the following attributes:

- **Exception Type:** Missing Evidence, Data Conflict, Authority Unresolved, or Skip Trace Exhaustion.
- **Priority Rating:** Critical (Equity $> \$200k$), High (Equity $\$100k$ – $\$200k$), Normal ($< \$100k$).
- **Resolution SLA:** Critical exceptions must be resolved within 4 hours; High/Normal within 24 hours.

3. QC Performance Metrics & Thresholds

METRIC	TARGET STANDARD	CURRENT BENCHMARK
Parcel Attribution Accuracy	$> 98.5\%$	99.1%
Fiduciary Authority Accuracy	$> 95.0\%$	96.4%
Contact Line Active Rate	$> 92.0\%$	94.2%
First-Pass Automated Yield	$> 75.0\%$	78.5%
Client-Reported Defect Rate	$< 2.0\%$	1.1%
Exception Resolution Turnaround	< 24 Hours	14.2 Hours

Client Delivery Workflow

Institutional delivery pipeline, multi-channel dispatch protocols, and disposition telemetry loop powering exclusive acquisition partnerships.



Exclusivity-First Delivery Paradigm

Gieni strictly rejects commodity list dumping. In each active county, Gieni operates under a single-partner exclusivity model. Opportunities are never co-brokered, shared, or recycled. Every delivery package represents a certified, decision-maker-ready acquisition file delivered via scheduled batch or instant Priority A flash alert directly into the partner’s CRM and secure data room.

1. Delivery Architecture & Dispatch Flow



2. The Four Specialized Delivery Channels



Channel 1: Scheduled Weekly Batch (Monday 08:00 AM Local)

- Cadence: Every Monday morning prior to acquisition team standup.
- Scope: Complete weekly harvest of reconciled probate filings across the partner's exclusive territory.
- Packaging: Curated database sync + structured weekly review packet ranked by Composite Score.
- Target SLA: Delivered with 100% QC certification and 0 unverified entities.



Channel 2: Urgent Priority A Flash Alerts (< 4 Hour Latency)

- Trigger Criteria: Composite Score ≥ 80 , estimated net equity > \$150,000, and Tier 1 or Tier 2 authority status.
- Dispatch: Immediate push via SMS dispatch to acquisitions lead + urgent email alert containing pre-compiled deal brief.
- Strategic Objective: Ensure partner initiates contact before competing investors or retail agents discover the docket filing.



Channel 3: Direct Bi-Directional CRM Integration

- Supported CRMs: Podio, Salesforce, HubSpot, RElsift, Follow Up Boss, and custom Webhooks.
- Mapping: Real-time creation of Lead, Property, Contact, and Opportunity records with mapped custom fields (e.g. Authority Tier, Net Equity, APN).
- Bi-Directional Sync: Dispositions logged in the client's CRM automatically sync back to Gieni OS to close the feedback loop.



Channel 4: Secure Data Room & Notion Partner Dashboard

- Access: Dedicated single-tenant Notion portal or encrypted document repository.
- Contents: Live interactive Kanban views, historical delivery archives, county playbook guides, and downloadable PDF evidence dossiers containing raw deeds, wills, and court dockets.

3. Anatomy of the Certified Opportunity File

Every delivered opportunity includes an institutional 8-point intelligence dossier designed to give acquisitions specialists immediate conversational dominance:

- **1. Executive Opportunity Brief:** Case Number, Filing Date, Viability Score (0–100), Priority Tier (A/B/C), and SLA Expiration.

- **2. Physical Property Profile:** APN/PIN, Situs Address, Legal Description, GIS Boundary Coordinates, Zoning Class, and Physical Condition Indicators.
- **3. Ownership & Equity Waterfall:** Assessed Value, Estimated AVM Market Value, Open Mortgages/Notes, Tax Delinquency, and Net Equity (\$ and %).
- **4. Control Dynamics & Archetype:** Identified De-Facto Decision-Maker, Heir Conflict Level, Occupancy Status (Vacant / Heir Occupied / Tenant), and Social Dynamics.
- **5. Authority Pathway & Signatory Capacity:** Four-Tier Authority Status, Letters Testamentary/Administration status, Court Power Scope (Independent vs. Dependent), and Probate Docket URL.
- **6. Verified Decision-Maker Contact Vector:** Validated Primary Phone Numbers (Mobile/Landline with DNC check), Current Mailing Address, Validated Email, and Attorney Quarantine status.
- **7. Action Directives & Empathy Playbook:** Recommended Outreach Channel, First-Touch Script Angle, Probate Timeline Context, and Legal Hurdles to anticipate.
- **8. Chain-of-Custody Evidence Dossier:** Direct hyperlinked PDFs of Probate Petition, Last Will & Testament, Recorded Conveyance Deed, and Assessor Tax Card.

4. Service Level Agreements (SLAs) & Defect Guarantee



Zero-Defect Commitment & Credit Policy

Gieni guarantees evidentiary accuracy. If a delivered file contains an erroneous APN match, a deceased fiduciary, or a property not part of the estate, the partner submits a one-click defect report. Gieni replaces the opportunity file or credits the partner account within 24 hours.

5. The Disposition Telemetry & Feedback Flywheel

Delivery is not the end of the workflow; it is the beginning of the learning loop. Partners log 5 standard telemetry milestones for each delivered file:

1. **Contact Attempted:** Timestamp and channel (Call, SMS, Direct Mail, Door Knock).
2. **Decision-Maker Connected:** Confirmation of authority and sentiment rating (Motivated, Neutral, Hostile).
3. **Appointment Scheduled:** Physical walk-through or virtual consultation date.
4. **Offer Submitted:** Cash offer amount, contract terms, and seller counter-offer.
5. **Contract Executed / Closed Won:** Purchase price, assignment/wholesale fee or projected equity spread.



Every disposition outcome directly feeds the Feedback Agent (FA), dynamically updating county friction coefficients and recalibrating the Opportunity Scoring Engine.

Data Architecture

Data Architecture Specification

Gieni Acquisition Decision Intelligence Platform

Version: 1.0 | Classification: Internal Platform Architecture | Owner: Data Platform | Status: Approved



1. Purpose & Mission

The Gieni Data Platform transforms fragmented public records into acquisition-ready intelligence by modeling: Properties, Parties, Estates, Trusts, Ownership, Control, Authority, Evidence, and Outcomes.

Unlike traditional lead-generation systems that rely on flat, denormalized spreadsheets, Gieni operates through a normalized relational and graph architecture designed to support: Ownership Intelligence, Control Intelligence, Authority Resolution, Opportunity Scoring, and Outcome Learning.

2. Architectural Principles



Principle 1: Property is not opportunity.

Property ≠ Opportunity

Physical real estate is merely an underlying asset. An acquisition opportunity only exists when equity spread, statutory authority, and seller motivation converge.



Principle 2: Ownership is not control.

Ownership ≠ Control

Title records reflect historical legal vesting. De-facto control is governed by physical occupancy, family dynamics, informal leadership, or probate administration.



Principle 3: Control is not authority.

Control ≠ Authority

A motivated heir may control the physical house, but only a court-appointed fiduciary with statutory Letters possesses the legal power to execute an enforceable deed conveyance.

**Principle 4: Every conclusion must be traceable.**

All intelligence outputs must be tied to: Evidence Artifact, Schema Version, Primary Source URL, Ingestion Timestamp, and Verifying Agent / Reviewer.

**Principle 5: Every assessment is temporal.**

The system must preserve what was known, when it was known, and how it changed over time as court dockets progress from petition to closing.

3. Platform Domains

Rather than a flat cluster of tables, Gieni partitions all platform entities into six bounded operational domains:

**Domain A: Jurisdiction Domain**

- Objects: County, Court, Recorder, Assessor, Source Adapter
- Purpose: Represents municipal systems and government portals that generate primary public records.

**Domain B: Identity Domain**

- Objects: Party, Person, Organization, Contact Point, Role Assignment
- Purpose: Represents people, legal entities, and fiduciaries appearing in municipal records, establishing canonical deduplicated identity profiles.

**Domain C: Legal & Property Domain**

- Objects: Probate Case, Estate, Property, Trust, Ownership Event, Asset Interest, Stakeholder Relationship
- Purpose: Represents legal ownership structures, real estate assets, and historical deed chains.

**Domain D: Intelligence Domain**

- Objects: Ownership Assessment, Control Assessment, Authority Assessment, Valuation Assessment, Equity Assessment, Opportunity Score
- Purpose: Stores synthetic outputs, confidence ratings, and decision directives produced by intelligence engines.



Domain E: Commercial Domain

- Objects: Opportunity, Opportunity Delivery, Client, Territory Agreement, Disposition
- Purpose: Connects verified intelligence with commercial partners, governing exclusivity boundaries and delivery tracking.



Domain F: Evidence Domain

- Objects: Evidence, Claim, Claim Evidence, Review Decision, Exception, Workflow Run
- Purpose: Provides auditability, immutable SHA-256 chain-of-custody tracking, and human-in-the-loop exception management.

4. Core Entity Specifications



County Entity

- Purpose: Represents a jurisdictional territory.
- Primary Key: county_id (UUID)
- Relationships:
 - County 1:M Probate Cases
 - County 1:M Properties
 - County 1:M Territory Agreements



Probate Case Entity

- Purpose: Represents a municipal court proceeding.
- Key Fields: case_id, case_number, filing_date, case_status, case_type, presiding_judge
- Relationships:
 - Probate Case 1:1 Estate
 - Probate Case 1:M Stakeholders
 - Probate Case 1:M Authority Candidates



Property Entity

- Purpose: Represents a physical parcel reconciled against county GIS and tax rolls.
- Key Fields: property_id, apn, situs_address, legal_description, property_type, county_id
- Relationships:
 - Property M:M Asset Interests
 - Property 1:M Ownership Events
 - Property 1:M Valuation Assessments

5. Graph Relationship Model

Graph edges are explicitly separated from static entity tables. The Relationship table stores weighted, contextual graph edges between any two nodes in the system:

Relationship Edge Schema

- Schema: relationship_id (UUID), from_entity (URI), to_entity (URI), relationship_type (Enum), confidence (Float 0–1.0), source_claim (FK → Claim.claim_id)
- Relationship Types:
 - Owns / Formerly Owned
 - Controls / Potential Controller
 - Represents / Trustee Of / Beneficiary Of
 - Heir Of / Connected To / Authority Candidate
- Graph Architecture: Makes the platform graph-native, allowing graph traversal algorithms to unmask non-obvious decision pathways.

6. Intelligence Layer (Decoupled Assessments)

Gieni strictly prohibits storing intelligence conclusions inside raw source entities. Assessments are stored as discrete, versioned records linked to their respective engines:



Ownership Assessment

- Produced By: Ownership Intelligence Engine (OIE)
- Outputs: Ownership Path (Sole / Joint / TIC / Trust), Ownership Complexity Score (10–100), Ownership Confidence Rating.



Control Assessment

- Produced By: Control Intelligence Engine (CIE)
- Outputs: Control Archetype (Unified / Informal Leader / Proxy / Trust Fiduciary / Contested), Primary Controller Person ID, Consensus Requirement Flag, Control Confidence Rating.



Authority Assessment

- Produced By: Authority Resolution Engine (ARE)
- Outputs: Authority Tier (Tier 1 Certified, Tier 2 Nominated, Tier 3 Intestate Consensus, Tier 4 Unresolved), Authority Confidence Rating, Statutory Power Scope (Full Power vs Court Confirmation Required).



Opportunity Assessment

- Produced By: Opportunity Scoring Engine (OSE)
- Outputs: Composite Viability Score (0–100), Priority Band (A / B / C), Deal Friction Score (DFS), Recommended Action Playbook.

7. Evidence Architecture: Claims & Verifiable Proof



The Claim-Evidence Triad

1. Claim: Represents an atomic fact, calculation, or inference.
 - Fields: claim_id, claim_type, value, confidence, timestamp
2. Evidence: Represents immutable primary source material.
 - Examples: Probate Petition PDF, Recorded Conveyance Deed, County Assessor Roll, Court Minute Order, Trust Instrument
3. Claim Evidence: Explicitly links evidence artifacts to claims.
 - Concrete Example:
 - Claim: "John Smith is probable fiduciary"
 - Supported By: Evidence [Letters Testamentary, Page 7, Paragraph 2]
 - Hash: SHA-256 verification string

8. Opportunity Delivery Layer (Separating Intelligence from Commerce)



Opportunity vs. Opportunity Delivery

- Opportunity: Internal intelligence synthesis record (asset, authority, score, dossier).
- Opportunity Delivery: The commercial event governed by county exclusivity.
 - Fields: delivery_id, client_id, opportunity_id, delivery_date, delivery_channel (Weekly Batch / Priority A Alert), capacity_counted, billing_state
- Strategic Impact: Completely isolates proprietary intelligence from partner licensing and billing logistics.

9. The 14-Stage Opportunity Lifecycle

- | | |
|------------------------|--|
| 1. Discovered | — Court docket scraped & immutable hash assigned |
| 2. Property Identified | — Assessor GIS parcel match (PAS ≥ 70) |
| 3. Ownership Resolved | — Title deed chain & Net Equity calculated |
| 4. Control Mapped | — Informal leader & decision archetype isolated |
| 5. Authority Resolved | — Court Letters & statutory power verified |

- | | |
|-----------------|--|
| 6. Scored | — Composite Viability Score & Priority A/B/C assigned |
| 7. QC Certified | — Passed deterministic 6-gate verification |
| 8. Delivered | — Synced to exclusive partner CRM & data room |
| 9. Contacted | — Partner outreach initiated |
| 10. Appointment | — Consultation / inspection scheduled |
| 11. Offer | — Formal acquisition offer submitted |
| 12. Contract | — Purchase contract executed |
| 13. Closed | — Title recorded & wholesale / equity fee realized |
| 14. Archived | — Lifecycle completed & telemetry committed to ML training |

10. Learning Layer & Continuous Feedback Loop



The Closed-Loop Disposition Flywheel

Disposition Telemetry (Contact, Appointment, Offer, Contract, Revenue)

↓

Feedback Agent (FA) / Ingestion Webhook

↓

Model Recalibration (Adjusting County Friction & Weight Coefficients)

↓

Updated Scoring Precision & Conversion Multipliers

11. Master System Architecture Pipeline

PUBLIC RECORDS (Court Dockets & Deeds)

|

▼

PROPERTY INTELLIGENCE

|

▼

OWNERSHIP INTELLIGENCE

|

▼

CONTROL INTELLIGENCE

|

▼

AUTHORITY INTELLIGENCE

|

▼

OPPORTUNITY INTELLIGENCE

|

▼

PROBATE OPPORTUNITY FILE

|

▼

CLIENT OUTCOMES

|



Gieni converts fragmented public records into acquisition-ready decisions through a verifiable, domain-driven, graph-native intelligence platform.

Complete Automation Map

End-to-end automation topology classifying ingestion, resolution, quality control, and dispatch workflows across autonomous agents and human specialists.

 **Automation Philosophy: Autonomous Execution with Deterministic Containment**

Gieni automates 85%+ of the end-to-end research and delivery lifecycle through scheduled scrapers, multi-agent LLM analysis, and event-driven webhooks. However, where legal title, fiduciary authority, or large capital allocations are concerned, full automation without human oversight creates catastrophic error rates. Gieni enforces a three-tier classification: Fully Automated, Semi-Automated (Human-in-the-Loop), and Pure Manual Specialist Operations.

1. Pipeline Automation Matrix

STAGE	AUTOMATION LEVEL	PRIMARY ENGINE / AGENT
1. Court Docket Scraping	[FULLY AUTOMATED]	Municipal Ingestion Worker
2. Property Matching & APN	[FULLY AUTOMATED]	Property Identity & Parcel Recon
3. Deed Retrieval & Equity	[FULLY AUTOMATED]	Ownership Intelligence Engine
4. Stakeholder & Control Parse	[SEMI-AUTOMATED]	Control Intelligence Agent
5. Authority Classification	[SEMI-AUTOMATED]	Authority Resolution Engine
6. Decision-Maker Enrichment	[FULLY AUTOMATED]	Enrichment Pipeline Worker
7. Viability Scoring (0-100)	[FULLY AUTOMATED]	Opportunity Scoring Engine
8. Quality Control Gate	[SEMI-AUTOMATED]	QC Agent & Human Reviewer
9. Client Delivery & CRM Sync	[FULLY AUTOMATED]	Reporting Agent & Webhook Dispatch
10. Disposition Feedback Loop	[SEMI-AUTOMATED]	Feedback Agent & Account Exec
11. County Playbook Expansion	[MANUAL SPECIALIST]	Gieni Expansion Taskforce

2. Fully Automated Workflows (Zero Human Touch)



Automated Court Scrapers & Docket Ingestion

- Mechanism: Scheduled headless browser workers (Playwright) trigger on a daily cron at 03:00 AM local time.
- Execution: Authenticates with municipal portals, queries probate docket registries for new filings, downloads petition PDFs, and extracts docket numbers into the Counties and Probate Opportunities databases.
- SLA / Runtime: Execution time < 15 minutes per county.



Parcel Attribution & GIS Property Matching

- Mechanism: Property Identity Agent (PIA) + Parcel Reconciliation Agent (PRA).
- Execution: Matches decedent name and last known domicile against County Tax Assessor GIS vector layers. Calculates the Parcel Attribution Score (PAS). If $PAS \geq 70$, the APN, situs address, and tax assessment are automatically committed.



Opportunity Scoring & Priority Tiering

- Mechanism: Opportunity Scoring Agent (OSA).
- Execution: Computes Composite Viability Score (0–100) using the multi-variable formula balancing Net Equity against Deal Friction Score (DFS). Automatically assigns Priority A (≥ 80), Priority B (60–79), or Priority C (40–59).



Multi-Channel Dispatch & Partner CRM Webhooks

- Mechanism: Reporting Agent (RA).
- Execution: Upon QC certification approval, RA dispatches real-time webhooks to the partner's CRM (Podio / Salesforce / HubSpot), triggers SMS flash alerts for Priority A deals, and stages weekly batch records.

3. Semi-Automated Workflows (Human-in-the-Loop)

In high-friction or high-ambiguity probate scenarios, autonomous agents perform initial heavy extraction and draft structured hypotheses, then route edge cases to human specialists:



Authority Resolution & Ambiguity Review

- **Agent Action:** Extracts petitioner, nominated executor, and Letters status from probate pleadings.
- **Human Trigger:** If docket indicates competing petitions, will contests, or dependent administration requiring court confirmation.
- **Specialist Role:** Probate analyst reviews scanned codicils and updates Authority Classification to Tier 1, 2, 3, or Unresolved.



Pre-Delivery Quality Control Certification

- **Agent Action:** Quality Control Agent (QCA) performs automated 10-point cross-reference validation.
- **Human Trigger:** Any score discrepancy, conflicting deed dates, or borderline Priority A files (scores 78–82).
- **Specialist Role:** QC Lead inspects the evidence pack and toggles QC Status to Certified or pushes to Tasks & Exceptions.



Monthly Model Recalibration & Weight Adjustments

- **Agent Action:** Feedback Agent (FA) aggregates partner disposition data (appointment rates, closed wholesale fees).
- **Specialist Role:** Lead Data Scientist reviews coefficient suggestions and recalibrates county weights.

4. Pure Manual Specialist Operations

- **County Feasibility Audits:** Initial manual reconnaissance of county clerk portals, court software types, and filing volumes before automated scraper deployment.
- **Partner Onboarding & Buy-Box Calibration:** Executive strategy sessions to define equity thresholds, asset criteria, and custom CRM field mapping.
- **Legal & Regulatory Compliance Audits:** Quarterly review of public records privacy guidelines, state solicitation statutes, and DNC registry compliance.

5. Core Automation Technology Stack



Production Infrastructure Stack

- Ingestion & Scraping: Playwright / Node.js headless workers deployed on AWS ECS with residential proxy rotation.
- Relational Storage: Managed PostgreSQL (Amazon RDS) with PostGIS extensions for spatial parcel queries.
- Orchestration & Queuing: Redis BullMQ for distributed task processing and Temporal for DAG workflow coordination.
- Operational Operating System: Notion REST API v2022-06-28 powering the Gieni OS workspace databases.
- Partner Integration: REST Webhooks, SendGrid transactional email, and Twilio SMS dispatch.

Opportunity Scoring Engine

Multi-variable prioritization engine that synthesizes equity spread, authority certainty, distress indicators, and Deal Friction Scores (DFS) into an objective 0–100 ranking and actionable acquisition priority bands.



Strategic Moat: Balancing Equity Spread Against Execution Friction

High equity is completely deceptive if execution is blocked by severe title friction or court gridlock. A \$400k equity spread locked behind 6 fighting heirs and court-supervised confirmation is far less valuable than a \$100k equity spread held by a confirmed sole executor ready to sign today. OSE integrates the Deal Friction Score (DFS) directly with financial upside to determine true probability-adjusted deal velocity.

1. Master Scoring Formula & Friction Calibration

The composite Opportunity Score is calculated by evaluating gross opportunity upside across four core pillars, directly modified by the calibrated Deal Friction Score (DFS):

$$\text{Gross Upside} = (\text{P1_Equity} * 0.35) + (\text{P2_Authority} * 0.30) + (\text{P3_Distress} * 0.20) + (\text{P4_Liquidity} * 0.15)$$

$$\text{Composite Score} = \text{Gross Upside} - \text{Deal Friction Score (DFS)}$$

Where DFS (0 - 35 deduction points) is computed from:

- Ownership Complexity (from OIE): Score 10 - 100
- Control Complexity (from CIE): Single controller vs. multi-heir consensus
- Court Oversight Level (from ARE): Independent Admin (0 pts) vs. Dependent/Supervised (-15 pts)
- Title & Debt Risk: Unprobated deeds, tax deed threats, or clouds on title

2. The Deal Friction Score (DFS) Architecture

DFS aggregates outputs generated across OIE, CIE, and ARE into an objective friction penalty that prevents high-equity/high-friction traps:



Deal Friction Scoring Matrix:

- Low Friction (0 - 5 pts penalty): Single owner, sole executor, independent administration, clean deed chain. [Fastest closing path]
- Moderate Friction (10 - 15 pts penalty): Trust transfer requiring certificate, or uncontested petition awaiting hearing. [Standard timeline]
- High Friction (20 - 25 pts penalty): 3+ heirs in intestacy, dependent court confirmation of sale required, or back-tax auction risk.
- Extreme Friction (30+ pts penalty): Multi-generational unprobated estate, active will dispute/caveat, or unlocated beneficiaries.



Strategic Calibration Rule:

- High Equity + Low Friction = Priority A (Immediate blitz outreach; target close in 14-21 days)
- High Equity + Extreme Friction = Priority B or C (High value, but long gestation cycle; assigned to specialized title-cure reps)

3. Priority Bands & Operational Directives



Priority A (Score 80 - 100) — Immediate Execution

- Profile: High equity spread (>50%), confirmed authority or sole petitioner, Low Deal Friction (DFS ≤ 10), motivated absentee or vacant asset.
- Protocol: Rapid 24-hour multi-channel outreach. Priority alerts to partner acquisitions director.



Priority B (Score 60 - 79) — Core Production Queue

- Profile: Strong equity with Moderate Friction, or moderate equity with Low Friction. Viable pipeline opportunity.
- Protocol: Structured 3-week outreach sequence.



Priority C (Score 40 - 59) — Nurture & Milestone Watch

- Profile: High friction (contested claims, multiple heirs, dependent court supervision). Requires milestone resolution before contract readiness.
- Protocol: Monitored by court docket state machine; re-scored upon milestone trigger.



Disqualified (Score < 40 or Hard Flag) — Suppressed

- Profile: Non-buybox asset, negative equity, unresolvable title defect, or DNC flag.
- Protocol: Auto-suppressed. Never charged against partner monthly capacity.

4. Dynamic Event-Driven Re-Scoring Engine

Scores automatically update in real time when upstream docket or title milestones reduce Deal Friction:

- **Letters Issued:** Authority certainty jumps to 100 pts and DFS drops by 15 pts, instantly elevating high-equity Priority B/C files into Priority A.
- **Heir Renunciation / Joinder Filed:** Multi-heir complexity score is reduced in OIE, lowering DFS by 10 pts.

Authority Resolution Engine

Evaluates the legal authority pathway associated with identified decision-makers. Consumes upstream ownership and control profiles to determine probable signatory capacity and transaction velocity under court oversight.



Strategic Moat: Probable Fiduciary Pathway Resolution

ARE does not perform raw real estate search or property discovery—those functions are resolved upstream by the Ownership Intelligence Engine (OIE). Instead, ARE consumes OIE's verified ownership structure, parcel attribution, and control pathways to evaluate the legal authority of the indicated parties under probate statutes and court dockets.

1. Upstream Data Consumption (OIE -> ARE Input Contract)

To eliminate redundant processing, ARE operates exclusively on verified inputs delivered by OIE and CIE. ARE evaluates the legal mechanism required to execute disposition based on the established title profile:

- **Estate-Owned Property (Sole Owner):** OIE establishes title in decedent; ARE evaluates docket status to determine if available records indicate active Letters Testamentary / Administration granting power of sale.
- **Trust-Held Property:** OIE identifies trust vesting; ARE evaluates the non-probate authority pathway associated with the indicated Successor Trustee, verifying that probate filings do not conflict with trust disposition powers.
- **Survivorship Vesting (JTWROS / TBE):** OIE establishes survivorship vesting; ARE confirms evidence suggesting title vests in the surviving co-tenant by operation of law, alerting acquisitions that probate representatives hold no disposition authority.
- **Multi-Heir / Intestate Estate:** OIE establishes tenancy-in-common heir structure; ARE evaluates whether an administrator has been appointed with statutory powers or if all fractional heirs must join in the conveyance.

2. Four-Tier Authority Classification Matrix

Every opportunity is classified into an evidentiary authority tier reflecting the strength of documentation regarding who can legally authorize the transaction:



Tier 1: Confirmed Authority (High Evidentiary Confidence)

- Evidentiary Standard: Available court dockets indicate Letters Testamentary or Letters of Administration have been formally issued.
- Probable Signer: Identified Personal Representative (Executor / Administrator).
- Operational Protocol: Direct outreach to fiduciary. Evidence indicates Independent Administration powers allowing direct contract execution without prior court confirmation.



Tier 2: Likely Decision Maker (Moderate Evidentiary Confidence)

- Evidentiary Standard: Records reflect an uncontested petition nominating an executor, or sole surviving statutory heir prior to formal letters issuance.
- Probable Signer: Nominated Executor or Primary Petitioner awaiting scheduled hearing.
- Operational Protocol: Early relationship building. Position partner to assist with logistical requirements so contract execution is primed upon appointment.



Tier 3: Stakeholder Requiring Verification (Multi-Party Evidence)

- Evidentiary Standard: Dockets and vital records suggest multiple heirs with equal statutory priority, co-fiduciaries, or intestate co-tenancy.
- Probable Signer: Multiple co-signing stakeholders or designated family consensus leader.
- Operational Protocol: Multi-stakeholder consensus strategy. Flagged for title-clearing coordination.



Tier 4: Authority Unresolved (Incomplete Evidentiary Pathway)

- Evidentiary Standard: Competing petitions filed, caveat against will recorded, creditor-initiated probate, or missing heirs.
- Probable Signer: Indeterminate from available public filings.
- Operational Protocol: Quarantined in Tasks & Exceptions queue. Not delivered to client until docket research establishes an actionable pathway.

3. Probate Docket State Machine & Supervision Levels

ARE tracks the judicial state machine to determine whether disposition requires formal court confirmation (Dependent Administration) or proceeds by fiduciary discretion (Independent Administration):

```
interface AREAuthorityEvaluation {  
    docketCaseNumber: string;  
    sourceOieParcelId: string;  
    indicatedSignerName: string;  
    indicatedFiduciaryRole: "Executor" | "Administrator" | "Successor Trustee" | "Surviving Joint Tenant";  
}
```

```

authorityClassification: "Confirmed Authority" | "Likely Decision Maker" | "Stakeholder Requiring Verification"
oversightType: "Independent Administration" | "Dependent Administration (Court Supervised)";
courtConfirmationRequired: boolean;
evidentiaryBasis: string; // e.g., "Letters Testamentary recorded Book 412 Pg 98"
evidenceConfidenceScore: number; // 0 - 100
attorneyBypassStatus: {
  attorneyName: string;
  firmName: string;
  fiduciaryDirectContactIsolated: boolean;
};
}

```

4. The Gatekeeper Bypass Protocol

Court cover sheets routinely list the petitioning law firm rather than the fiduciary. ARE isolates direct decision-maker metadata while quarantining attorney contact information:

- **Extract Fiduciary Entity:** Extracts full legal name and residence of the personal representative from petition exhibits.
- **Quarantine Attorney Records:** Classifies attorney telephone, email, and law firm as legal reference data, preventing automated marketing from spamming estate counsel.
- **Direct Consumer Enrichment:** Routes fiduciary profile through consumer trace pipelines to isolate personal mobile and residence records.

5. Evidentiary Standards & Legal Boundary Disclaimer



Evidentiary Compliance Framework

All ARE determinations represent research findings and confidence-based inferences drawn from public court dockets. Gieni does not supply legal opinions or title guarantees. Partner agreements mandate that clients conduct independent legal and title review prior to closing.

Ownership Intelligence Engine

Master orchestration layer that determines the ownership-control pathway required to convert a probate-related property into an acquisition opportunity.



Strategic Moat: From "Finding Properties" to Ownership-Control Pathways

Most probate data products operate under a lead-generation paradigm: find a name in court records and sell it as a lead. Gieni operates under an acquisition intelligence paradigm. Real estate cannot be purchased from a "probate record"—it can only be acquired by resolving the legal ownership structure, mapping who holds actual disposition control, and establishing an actionable title pathway.



Why This Matters: The Five Questions That Define a Real Deal

Most probate investors buy lists and waste 90% of their outreach time.

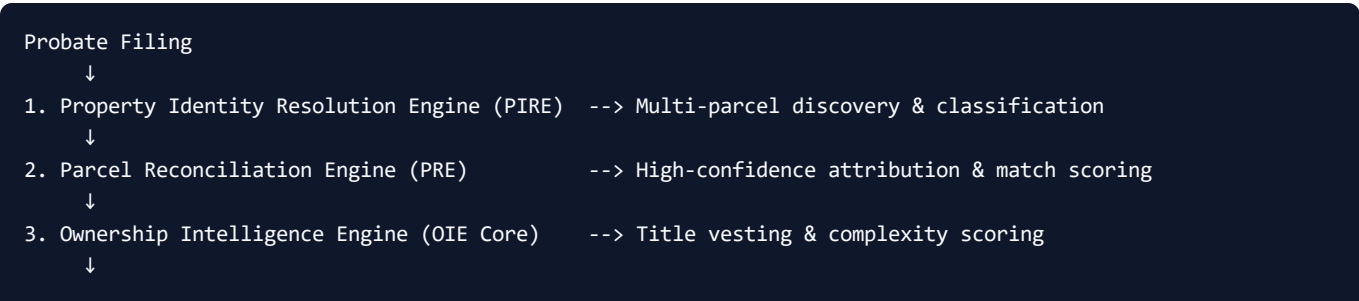
Gieni answers five core questions before an opportunity is ever delivered:

- 1. Which exact property belongs to the estate?
- 2. Who actually controls the disposition?
- 3. How much real, protected equity exists?
- 4. What title or heir complexity must be cured?
- 5. Does probate actually affect the legal ability to sell?

Only after answering all five questions does an opportunity become acquisition-ready.

1. The Master Pipeline Architecture

OIE orchestrates five specialized, patentable-grade submodules that convert raw courthouse filings into verified, decision-maker-ready opportunity files:



4. Control Intelligence Engine (CIE)	--> Ownership ≠ Control pathway separation
↓	
5. Equity Intelligence Engine (EIE)	--> Net equity waterfall & encumbrance audit
↓	
Authority Resolution Engine (ARE)	--> Fiduciary powers & state machine
↓	
Opportunity Scoring Engine (OSE)	--> 0-100 Priority ranking & SLA assignment

2. Submodule 1: Property Identity Resolution Engine (PIRE)

Probate dockets rarely map 1:1 to a single suburban home. A single decedent frequently owns multiple parcels, unbundled acreage, fractional interests, or assets under distinct legal addresses. PIRE answers the initial portfolio discovery questions:

- **Estate Asset Enumeration:** Determines the total count of real property assets owned by the decedent across the county and adjacent jurisdictions.
- **Asset Category Classification:** Distinguishes between Primary Homestead, Rental/Tenant-Occupied Asset, Vacant Investment Land, or Contiguous/Adjacent Tax Parcels.
- **Multi-APN Bundle Resolution:** Identifies cases where a single physical home sits across two separate tax parcels (e.g., house on Lot 1, garage/driveway on Lot 2), preventing catastrophic single-parcel contracts.

3. Submodule 2: Parcel Reconciliation Engine (PRE) & Attribution Score

Gieni rejects absolute claims of "100% match." Instead, PRE delivers high-confidence parcel attribution based on multi-source cross-validation and a proprietary 0–100 Parcel Attribution Score:

Parcel Attribution Score (PAS) Weights:

- Source Agreement (Assessor + Recorder + Court Petition): 30%
- Name Match Confidence (Jaro-Winkler string distance): 25%
- Address Correlation (Situs vs. residential voter/death): 20%
- Title Continuity (Unbroken deed recording history): 15%
- Tax Record Alignment (Current taxpayer of record): 10%

Scoring Thresholds:

- | | | |
|------------|------------------------|--|
| [90 - 100] | Verified Parcel Match | --> Certified for Automated Pipeline |
| [70 - 89] | Probable Parcel Match | --> Delivered with Validation Tag |
| [< 70] | Manual Review Required | --> Routed to Exception Queue (Tasks & Exceptions) |



Evidence Standard: Parcel match confidence must exceed the minimum delivery threshold (PAS ≥ 70) established by Gieni quality standards. Any case below 70 is flagged for human researcher verification.

4. Submodule 3: Ownership Complexity Engine (OCE)

OIE calculates an objective Ownership Complexity Score (10–100) reflecting the title friction required to convey marketable fee-simple title. This score feeds directly into Opportunity Scoring:

- **Score 10 (Minimal Friction):** Single Owner / Sole Fee Simple. Decedent owned 100% of the property. Clear testamentary or statutory path.
- **Score 35 (Low Friction):** Joint Ownership (JTWROS / TBE). Property transfers automatically by operation of law to the surviving co-owner outside of probate.
- **Score 45 (Moderate Friction):** Trust Ownership. Non-probate disposition. Requires locating the trust instrument and validating the Successor Trustee.
- **Score 65 (High Friction):** Multiple Heirs / Tenancy in Common (TIC). Intestate estate with multiple heirs holding fractional undivided interests.
- **Score 85 (Severe Friction):** Multiple Estates / Ancestral Heir Property. Title remains in the name of deceased ancestors, requiring serial probate or quiet title actions.
- **Score 100 (Maximum Friction):** Entity + Trust Stack. Complex layered holding structures (e.g., LLC owned by a Living Trust with a deceased managing member and contested probate).

5. Heir Property Models & Resolution Framework

Heir property represents one of the largest untapped opportunities in distressed acquisitions, but carries high execution risk without workflow segmentation. Gieni classifies heir properties into six structural models and four resolution difficulty tiers:



Heir Property Typology:

1. Single Estate: Clean decedent title, intestate with 2–4 immediate heirs. [Resolution: Easy]
2. Multi-Generation Estate: Title in grandparents; requires tracing lineage through deceased children. [Resolution: Complex]
3. Clouded Title: Unrecorded deeds, wild deeds, or ambiguous boundary descriptions. [Resolution: Specialist Review]
4. Missing Heirs: Known living heirs whose whereabouts are unknown. [Resolution: Moderate]
5. Unknown Heirs: Intestate succession requiring genealogical heir searches and affidavits of heirship. [Resolution: Complex]
6. Multiple Probate Chains: Interlocking probates across multiple counties or states. [Resolution: Specialist Review]

6. Submodule 4: Control Intelligence Engine (Ownership ≠ Control)

This distinction is Gieni's most defensible intellectual property moat. In probate real estate, the legal owner of record is frequently NOT the person who holds the legal power to convey title.



The Ownership vs. Control Matrix:

- Owner: Estate of Decedent --> Control: Court-Appointed Personal Representative (Executor / Administrator)
- Owner: Revocable Living Trust --> Control: Successor Trustee (Bypasses probate court entirely)
- Owner: 6 Surviving Heirs --> Control: Family Consensus Leader / Co-signing Heirs
- Owner: LLC or Partnership --> Control: Managing Member / Surviving General Partner
- Owner: Surviving Joint Tenant --> Control: Sole Living Co-Owner (Executor has zero authority)

By isolating Ownership from Control, OIE establishes the foundational data contract that directly feeds the Authority Resolution Engine (ARE). While OIE solves "What is the property and who holds title?", ARE solves "Who has legal signing authority to sell it?"

7. Submodule 5: Equity Intelligence Engine (EIE)

Simple "estimated equity" models fail because they ignore estate-specific liens and statutory claims. EIE computes an audited Net Equity Waterfall:

Net Equity Waterfall Formula:

```
Gross Market Value (Conservative AVM / Assessor baseline)
- (Estimated Open Mortgage Balance via original recording + amortization)
- (Delinquent Real Property Taxes + accumulated penalties & interest)
- (Municipal Code Enforcement, Weed, & Demolition Assessment Liens)
- (Federal Tax Liens & State Department of Revenue Liens)
- (Medicaid Estate Recovery Program [MERP] Priority Statutory Liens)
- (Estimated Statutory Probate Court Fees & Fiduciary Commissions)
- (Recorded Creditor Claims Against the Estate)
=====
= Net Actionable Estate Equity
```

Outputs are categorized into four Equity Tiers: Exceptional Equity (>70% net spread), High Equity (50%–70%), Moderate Equity (30%–50%), and Low/Negative Equity (<30% or underwater), which directly informs cash offer capacity.

8. Ownership Investigation Exceptions Engine

Robust automation requires deterministic exception handling. The Exception Engine automatically halts automated processing and diverts anomalous cases to the Manual Investigation Queue (Tasks & Exceptions

DB) upon encountering seven specific triggers:

- **Conflicting Deeds of Record:** Simultaneous or contradictory deed recordings within 180 days of death.
- **Multiple Non-Contiguous APNs:** Estate holds 3+ distinct properties requiring allocation decisions.
- **Unrecorded Trust Reference:** Deed conveys to a trust but no certification of trust or docket notice is found.
- **Missing Probate / Broken Deed Chain:** Gap between recorded deed grantee and decedent name.
- **Generational Name Collision:** Father and son sharing exact name (Sr./Jr.) without generational suffixes on title.
- **Out-of-State / Foreign Ownership:** Ancillary probate requirements in other state jurisdictions.
- **Dissolved Entity Ownership:** Property titled under an LLC that was forfeited or administratively dissolved prior to death.

9. Complete OIE Delivery Data Contract

Every completed opportunity delivered by Gieni carries an exhaustive, five-dimension property intelligence profile:



OIE Certified Opportunity Package Schema:

- 1. Property Profile: Situs Address, County, APN/PIN, Legal Lot/Block, Living Area, Lot Size, Land Use Code
- 2. Ownership Profile: Current Vesting of Record, Recorded Deed Instrument #, Deed Type, Ownership Structure, Complexity Score (10-100)
- 3. Equity Profile: Estimated Market Value, Assessor Value, Open Mortgages, Tax Delinquency, Lien Exposure, Net Equity Tier
- 4. Authority Inputs: Identified Fiduciaries, Trustees, Surviving Joint Tenants, Primary Potential Signers (Transferred to ARE)
- 5. Risk Profile: Title Risk Rating, Probate Dependency Rating, Transfer Friction, Marketability Score

Agent Architecture

Autonomous multi-agent orchestration architecture that converts raw municipal archives into deterministic, decision-maker-ready acquisition opportunities.



Master Orchestrator: Autonomous DAG Execution Layer

Gieni operates as a directed acyclic graph (DAG) of nine specialized autonomous agents overseen by the Master Orchestrator. The orchestrator enforces strict data contracts, manages asynchronous event streams, dispatches tasks, and routes edge cases to human-in-the-loop review queues when confidence scores breach minimum delivery thresholds.

1. Master Agent Interaction Diagram



2. The Nine Specialized Autonomous Agents



Agent 1: Property Identity Agent (PIA)

- Purpose: Discovers all physical real estate holdings belonging to a decedent estate and performs initial portfolio classification.
- Inputs: Probate petition text, decedent name, last known residence, tax ID aliases.
- Outputs: Candidate parcel inventory, asset type tags (Homestead, Rental, Acreage, Contiguous Lot).
- Dependencies: County court petition scrapers.
- Human Review: Triggered if estate holds 3+ non-contiguous properties or unaddressed parcels.
- KPIs: Parcel identification coverage (target >85%), false negative rate (<3%).
- Failure Modes: Common name collisions; fallback to probate petition asset schedules and voter registry addresses.



Agent 2: Parcel Reconciliation Agent (PRA)

- Purpose: Establishes high-confidence attribution between decedent identity and specific county tax assessor GIS parcel records.
- Inputs: Candidate parcels from PIA, county GIS layers, tax assessor rolls.
- Outputs: Matched APN/PIN, legal description, situs address, Parcel Attribution Score (PAS, 0–100).
- Dependencies: Property Identity Agent.
- Human Review: Required if PAS falls below delivery threshold (<70 pts).
- KPIs: Attribution accuracy (target >95%), unbundled parcel detection rate (>98%).
- Failure Modes: Out-of-county property ownership; fallback to statewide grantor/grantee index.



Agent 3: Ownership Intelligence Agent (OIA)

- Purpose: Audits deed history to identify legal title vesting, encumbrance liabilities, and Net Equity.
- Inputs: Reconciled APN, county recorder conveyance deeds, open mortgage notes, tax lien records.
- Outputs: Vesting classification (Sole Fee Simple, JTWROS, Trust, TIC, Heir Property), Ownership Complexity Score (10–100), Net Equity Waterfall.
- Dependencies: Parcel Reconciliation Agent.
- Human Review: Mandatory for multi-generational unprobated deeds or unrecorded trust instruments.
- KPIs: Deed instrument retrieval rate (>98%), net equity estimation precision ($\pm 10\%$).
- Failure Modes: Unrecorded deeds or wild deeds; fallback to title company plant index.



Agent 4: Control Intelligence Agent (CIA)

- Purpose: Decouples ownership from decision-making; isolates the de-facto human controller driving property disposition.
- Inputs: Heirs schedules, probate petition petitioner data, physical occupancy indicators, utility account metadata.
- Outputs: Control Archetype (Unified, Informal Leader, Proxy, Trust Fiduciary, Contested), primary decision-maker dossier.
- Dependencies: Ownership Intelligence Agent.
- Human Review: Triggered by hostile family caveats or competing sibling petitions.
- KPIs: Correct decision-maker attribution rate (>85%), outreach adoption rate (>80%).
- Failure Modes: Estranged multi-heir silence; fallback to geographical proximity heuristic.



Agent 5: Authority Resolution Agent (ARA)

- Purpose: Determines the legal authority pathway and statutory signatory capacity under probate court dockets.
- Inputs: Court minute orders, Letters Testamentary/Administration, will codicils, estate attorney records.
- Outputs: Four-Tier Authority Classification (Tiers 1–4), court oversight model (Independent vs. Dependent), gatekeeper-bypassed contact profiles.
- Dependencies: Control Intelligence Agent & Court Docket Feeds.
- Human Review: Required for dependent administration sales requiring court confirmation.
- KPIs: Authority accuracy rate (>90%), attorney quarantine success rate (>98%).
- Failure Modes: Court docket scanning lag; fallback to automated clerk calendar scraper.



Agent 6: Opportunity Scoring Agent (OSA)

- Purpose: Synthesizes financial upside against the Deal Friction Score (DFS) to assign an objective 0–100 viability ranking.
- Inputs: Net Equity tier, Authority certainty score, distress multipliers, Ownership Complexity score.
- Outputs: Composite Score (0–100), Priority Band (Priority A, B, C, Disqualified), actionable SLA assignment.
- Dependencies: Ownership Intelligence Agent, Authority Resolution Agent.
- Human Review: Edge cases scoring within ± 2 points of the Priority A threshold (78–82 pts).
- KPIs: Priority A deal conversion rate (>30%), disqualification accuracy (>99%).
- Failure Modes: Incomplete debt data; fallback to conservative maximum encumbrance default.



Agent 7: Quality Control Agent (QCA)

- Purpose: Acts as the pre-delivery gatekeeper, executing multi-point deterministic verification before file certification.
- Inputs: Assembled opportunity records from Agents 1–6.
- Outputs: QC Certification Pass/Fail, audit flag notes, routing to Tasks & Exceptions database.
- Dependencies: Agents 1–6.
- Human Review: Any file failing automated integrity checks (e.g. conflicting deed dates, mismatched parcel address).
- KPIs: Delivery pass rate (>95%), customer-reported defect rate (<2%).
- Failure Modes: Source data conflict; fallback to secondary title plant search.



Agent 8: Reporting Agent (RA)

- Purpose: Compiles certified files into standardized delivery packages and syncs to partner CRM platforms.
- Inputs: QC-passed opportunity records.
- Outputs: Weekly Batch Packages, Urgent Priority A Alerts, partner CRM record creation (Podio, Salesforce, HubSpot).
- Dependencies: Quality Control Agent.
- Human Review: CRM field mapping errors or delivery failure alerts.
- KPIs: Delivery SLA adherence (>99%), webhook dispatch latency (<30s).
- Failure Modes: Partner CRM API rate limits; fallback to encrypted portal drop and SMS alert.



Agent 9: Feedback Agent (FA)

- Purpose: Closes the intelligence loop by ingesting partner disposition telemetry and recalibrating scoring models.
- Inputs: Partner disposition tracking (contacts, appointments, offers, executed contracts, rejections).
- Outputs: Score weight recalibration factors, county-level friction adjustment coefficients, dynamic re-scoring triggers.
- Dependencies: Client Delivery & Disposition Tracking.
- Human Review: Periodic review with Gieni operations team during monthly strategy sessions.
- KPIs: Weekly disposition compliance (>80%), model predictive accuracy gain (>5% per quarter).
- Failure Modes: Missing partner feedback; automated reminders and partner check-in scheduling.

3. Architectural Guardrails: Deterministic Human-in-the-Loop

Gieni enforces a strict boundary between automated agent heuristics and human judgment. Autonomous agents perform 90%+ of data ingestion, reconciliation, and calculations, but exceptions are deterministically

isolated:



Autonomous agents propose; deterministic rules govern; human experts review exceptions; verified outcomes calibrate the next generation of models.

Customer Journey

The end-to-end operational lifecycle: transforming qualified real estate operators into exclusive, high-retention county partners through measurable acquisition outcomes.



Strategic Focus: Partnerships, Not Subscriptions

Gieni does not operate a self-serve SaaS portal. We become an exclusive, outsourced intelligence department for one operator per county. The customer journey is engineered to protect partner capacity, validate throughput in a paid 90-day sprint, and scale into multi-year recurring territory contracts.

1. Ideal Customer Profile (ICP)

Territory exclusivity is granted only to operators who possess the infrastructure to act rapidly on delivered opportunities:

- **Active Residential Wholesaler:** Closing 2–10+ distressed transactions per month with established title and escrow relationships.
- **Dedicated Acquisitions Team:** Trained acquisitions reps capable of initiating high-touch outreach within 24–48 hours of file delivery.
- **Distressed Property Investor / Fund:** Capable of structuring creative deal terms (probate advance, cash buyout, closing cost coverage) to solve seller hurdles.



Disqualified Buyer Profiles: Passive investors, retail realtors, casual house hunters, and solo operators lacking the capacity to contact sellers and make cash offers. Exclusivity is revoked if leads are not actively worked.

2. Awareness Stage: Problem Recognition

The prospect experiences acute operational pain from the commodity lead model:

- **Marketing Burn:** Spending \$4,000–\$10,000/mo on bulk direct mail and cold callers with response rates under 1%.
- **Rep Fatigue:** Acquisitions reps waste 90% of their day hitting attorney gatekeepers, estranged relatives, or dead-ends with no equity.

- **Competitive Pressure:** Frustration that generic probate lists are sold simultaneously to 30 other local wholesalers.
-

3. Sales Stage: Discovery & County Qualification

The sales cycle focuses on territory qualification rather than standard software demos:

- **Initial Strategy Session:** Review of current pipeline volume, target acquisition criteria, and local acquisitions capacity.
 - **County Availability Verification:** Audit of Gieni's Counties register to confirm territory exclusivity is unclaimed in target county.
 - **Buy-Box Discovery:** Capturing property types (SFR, 2–4 unit), target zip codes, price ceilings, and minimum spread expectations.
-

4. Proposal Stage: Audit & Founding Partner Terms

Gieni presents a tailored proposal grounded in local historical volume data:



Founding Partner Proposal Deliverables:

- **County Volume Audit:** 90-day baseline of qualified probate filings and real property match rates.
 - **Monthly Capacity Commitment:** Guaranteed delivery of completed files (e.g., 40 files/month for Founding Partner).
 - **Commercial Terms:** \$3,000/mo + \$2,000 one-time setup fee under a 90-day validation agreement.
 - **Strict Territorial Lock:** Full exclusivity covenant preventing competitor delivery in named county.
-

5. Onboarding Stage: Workflow Calibration (Days 1–15)

The 15-day setup sprint configures the local pipeline infrastructure before volume delivery:

- **Source Mapping:** Ingestion feeds connected to county court dockets, assessor rolls, and recorder deed books.
 - **Scoring Calibration:** Encoding partner-specific buy-box rules and Deal Friction Score (DFS) thresholds into OSE.
 - **Calibration Batch Review:** Delivery of 10–15 sample files to review evidence standards directly with the partner's acquisitions lead.
-

6. Delivery Stage: Production Cadence (Days 16–75)

Weekly batch delivery paired with event-driven priority triggers:

- **Scheduled Weekly Batches:** Delivered directly into partner CRM or secure portal every week (approx. 10 files/week).
- **Urgent Priority A Alerts:** Immediate notification upon court milestone events (e.g. Letters issued on a high-equity vacant home).
- **Weekly Disposition Reporting:** Partner updates contact outcomes, decision-maker accuracy, and offers made via shared tracker.

7. Outcome Stage: Conversion Milestones

Success is measured by downstream transaction milestones, not data volume delivered:



Key Outcome Progression:

1. Verified Decision-Maker Contact: Target 80%+ reach rate with controlling parties.
2. In-Person / Virtual Seller Appointments: Leading conversion signal of deal relevance.
3. Cash Acquisition Offers Submitted: Immediate validation of pricing alignment.
4. Executed Purchase Contracts: North Star metric defining partner ROI.

8. Renewal Stage: 90-Day Validation Review (Days 76–90)

At Day 90, Gieni and partner review the comprehensive scorecard to assess commercial performance:

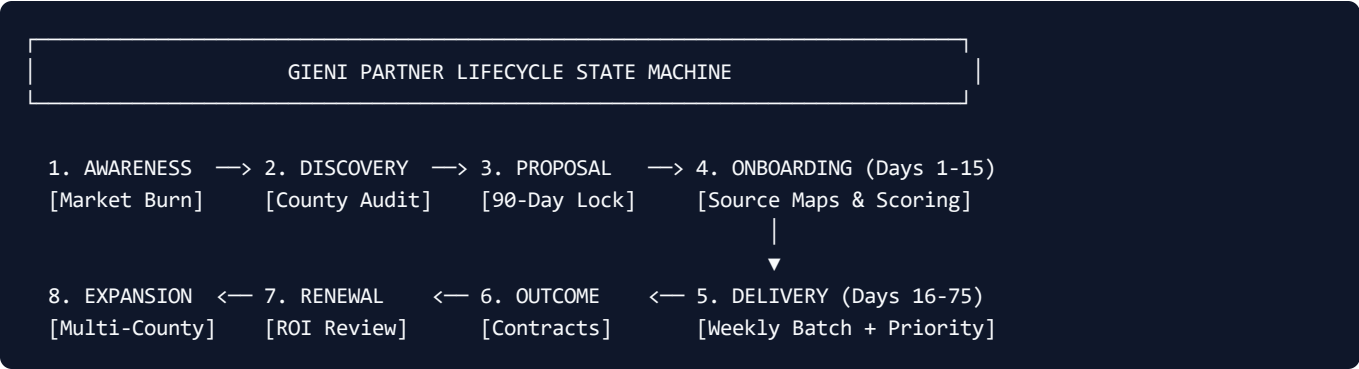
- **Attributed ROI Audit:** Comparison of closed contract assignment/spread revenue against total Gieni fees paid.
- **Contract Transition:** Transitioning from 90-day validation into 6-month or 12-month recurring exclusivity at locked partner rates.
- **Capacity Tier Adjustment:** Upgrading to County Growth (50 files/mo) or County Pro (85 files/mo) based on county throughput.

9. Expansion Stage: Regional Territory Dominance

Once county unit economics are established, partners systematically scale across surrounding markets:

- **Adjacent County Add-On:** Activating contiguous counties at 50% discount on standard setup fee.
- **Metro-Exclusive Lock:** Securing multi-county regional exclusivity to establish total off-market probate monopoly.

10. Complete Customer Journey Lifecycle Map



Competitive Moat

Gieni's competitive advantage is not access to data. It is the ability to convert fragmented ownership information into acquisition-ready decisions.



Strategic Moat: Decision Intelligence vs. Commodity Lead Lists

The real estate industry is flooded with data brokers who scrape public filings and sell identical spreadsheets to dozens of competing wholesalers. This commodity model destroys partner ROI. Gieni protects partners through territory exclusivity, proprietary decision engines, and a self-reinforcing outcome flywheel.

1. Industry Overview: The Commodity Data Trap

Traditional probate data vendors operate as low-margin data extractors. Their product catalog consists of:

- **Raw Court Scrapes:** Unfiltered case numbers and decedent names pulled without property verification.
- **Automated Skip-Tracing Dumps:** Spreadsheets containing 8 unverified phone numbers and 5 outdated email addresses per lead.
- **Non-Exclusive Reselling:** The exact same records are sold simultaneously to 20–50 competing investors in the same market.

2. Why That Model Fails: The Five Fundamental Unknowns

When an acquisitions team buys a traditional list, they do not receive acquisition opportunities. They receive homework. The team still does not know:

- **Which property matters?** Does the decedent own real estate? Which parcel belongs to the estate? Is it a home or a sliver of land?
- **Who owns it?** Is title held in fee simple, an unrecorded trust, survivorship tenancy, or an unprobated ancestral deed?
- **Who controls the decision?** Who holds physical custody, lives in the house, or serves as the informal family consensus leader?
- **Who can legally sign?** Does the petitioner hold letters of administration, or are 6 estranged siblings required to execute the deed?

- **Is the deal worth pursuing?** Is equity preserved, or will the transaction collapse under reverse mortgages, back taxes, and statutory liens?



The Economic Consequence: Acquisitions reps spend 90%+ of their working hours playing courthouse detective and cold-calling confused relatives rather than negotiating contracts with qualified decision-makers.

3. The Gieni Intelligence Stack

Gieni replaces manual courthouse detective work with a synchronized four-engine intelligence stack:

- **1. Ownership Intelligence Engine (OIE):** Reconciles real property parcels, title vestings, complexity scores (10–100), and net equity waterfalls.
- **2. Control Intelligence Engine (CIE):** Isolates de-facto decision dynamics, family hierarchy, resident caretakers, and control archetypes.
- **3. Authority Resolution Engine (ARE):** Evaluates legal signatory capacity, court oversight levels, and executes the gatekeeper bypass protocol.
- **4. Opportunity Scoring Engine (OSE):** Balances equity upside against the Deal Friction Score (DFS) to assign calibrated priority bands (A, B, C).

4. The Conceptual Core: Ownership ≠ Control ≠ Authority

The cornerstone of Gieni's defensibility is the systematic decoupling of the three layers that govern transitional real estate:



The Triad in Practice:

- Scenario A: Owner = Estate | Control = Caretaker Daughter | Authority = Out-of-State Executor
--> Strategy: Connect with the daughter to address maintenance and moving needs; execute contract through executor.
- Scenario B: Owner = Revocable Trust | Control = Successor Trustee | Authority = Successor Trustee
--> Strategy: Complete non-probate transaction; execute private sale under trust powers.
- Scenario C: Owner = 6 Heirs in Intestacy | Control = Eldest Brother | Authority = Unresolved
--> Strategy: Secure consensus with family leader first; coordinate estate administration to clear title.

5. The Result: Proprietary Information Asymmetry

When competitors attempt outreach on a probate lead, they sound like telemarketers: "Hi, do you own the property on 4th Street?" They are met with hostility and immediate rejection.



The Gieni Partner Advantage:

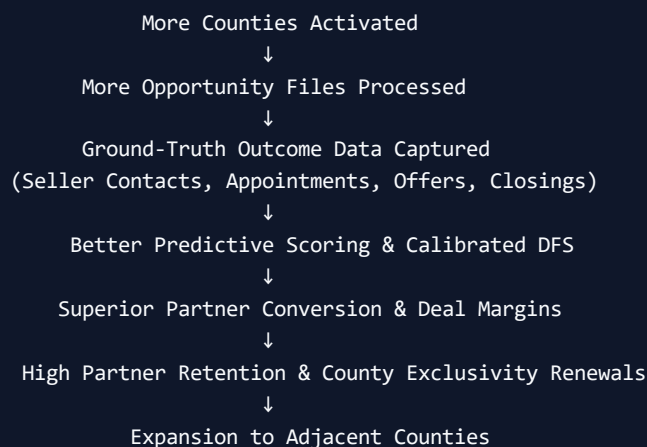
A Gieni partner enters the exact same situation knowing:

1. The home has \$180,000 in net equity with no open mortgage.
2. The deed is titled in the deceased mother's name.
3. The son living in the property is the primary caregiver seeking a stress-free transition.
4. Letters of Administration were issued with Independent Powers two weeks ago.

The partner speaks directly to the seller's real problem, structuring an acquisition before competitors even determine who lives in the home.

6. The Gieni Flywheel

Gieni's defensibility compounds over time through a closed-loop data and outcome flywheel:



7. The Four Long-Term Moat Categories



1. The Data Moat

- Deep jurisdictional source maps across 3,000+ counties.
- Historical docket milestone histories that track case progression over time.
- Proprietary cross-indexing between court dockets, GIS parcels, and deed books.



2. The Workflow Moat

- Institutional-grade research protocols and quality control gates.
- Automated exception-handling engines routing anomalies to research queues.
- Attorney gatekeeper bypass algorithms isolating direct decision-maker metadata.



3. The Decision Moat

- Mathematical Deal Friction Scoring (DFS) preventing high-equity traps.
- Algorithmic heuristics that identify de-facto family controllers before outreach.
- Title complexity categorization mapping exact legal pathways to closing.



4. The Outcome Moat

- Real-world disposition feedback captured weekly from county partners.
- Attribution loops connecting specific docket attributes to closed acquisition contracts.
- Continuous model calibration that public scrapers and data brokers cannot replicate.

8. Conclusion



Gieni's competitive advantage is not access to data. It is the ability to convert fragmented ownership information into acquisition-ready decisions.

Executive System Architecture

Convert fragmented public records into evidence-backed, decision-maker-ready acquisition opportunities.



Category Definition: Acquisition Decision Intelligence (ADI)

Gieni is not a lead-generation company, a data broker, or a skip-tracing service. Gieni is an Acquisition Decision Intelligence platform. Legacy vendors sell spreadsheets of unstructured names scraped from courthouses. Gieni transforms fragmented, multi-jurisdiction public archives into deterministic ownership-control pathways and actionable, contract-ready acquisition packages.



The Strategic Foundation: Ownership ≠ Control ≠ Authority

Traditional probate lists fail because they conflate three entirely distinct legal and human dimensions into a single row:

- The Owner of record is dead (or a trust entity).
- The Control belongs to a family member, caretaker heir, or resident occupant.
- The Authority belongs to a court-appointed fiduciary granted statutory Letters.

Only when an intelligence platform systematically resolves all three layers does a raw public filing become a viable acquisition opportunity.

1. Problem Statement: Generational Transfer Friction

Generational property transfer represents one of the largest ownership transitions in modern real estate. The vast majority of this real estate passes through complex, highly frictional public bureaucracies: county probate dockets, deed registries, tax assessor offices, and municipal courts.

For professional real estate wholesalers, acquisitions teams, and institutional buyers, sourcing off-market transactions in this sector is plagued by extreme operational waste:

- **Information Fragmentation:** Courthouse dockets do not list property addresses; deed records do not reflect death events; assessor rolls do not reflect fiduciary appointments.
- **Wasted Outbound Marketing:** Over 85% of generic marketing touches target individuals who cannot legally sign a deed—such as unappointed petitioners, estranged heirs, or attorney gatekeepers.

- **Title Gridlock & Delay:** Deals collapse weeks into escrow due to unresearched encumbrances, court-supervised confirmation delays, or unlocated co-heirs.

2. Why Now?



The Timing Imperative

Court records are public.

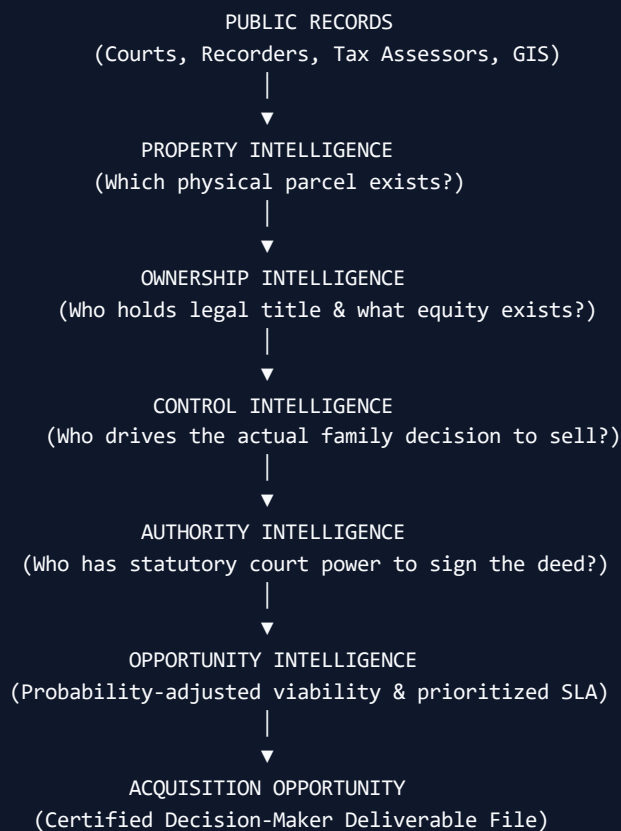
Property records are public.

GIS records are public.

Tax records are public.

The information has always existed. What has changed is the ability to reconcile fragmented ownership, control, authority, and equity at scale through autonomous multi-agent reasoning and deterministic relational modeling.

3. The Gieni Core Intelligence Pipeline



4. The Five Intelligence Layers Explained



Layer 1: Property Intelligence (PIRE & PRE)

- Core Question: Which exact real estate parcel is tied to this proceeding?
- Mechanics: Resolves fuzzy name variations, multi-parcel estate holdings, contiguous tax parcel bundles, and situs vs. mailing address discrepancies.
- Proprietary Metric: Parcel Attribution Score (PAS, 0–100) combining source agreement, address correlation, and title continuity.



Layer 2: Ownership Intelligence (OIE Core & EIE)

- Core Question: Who owns the property and what encumbrances exist?
- Mechanics: Analyzes recorded conveyance deeds to verify vesting (Fee Simple, JTWROS, Living Trust, Tenancy in Common, Heir Property). Computes the Net Equity Waterfall auditing open mortgages, tax arrears, and statutory liens.
- Proprietary Metric: Ownership Complexity Score (10–100) & Net Equity Tier.



Layer 3: Control Intelligence (CIE)

- Core Question: Who drives the family decision to sell?
- Mechanics: Disentangles Ownership from Control. Identifies informal family leaders, resident caretaker heirs, and proxy controllers using occupancy, legal retainer sponsorship, and geographic proximity heuristics.
- Proprietary Output: Control Archetype (Unified, Informal Leader, Proxy, Trust Fiduciary, Contested).



Layer 4: Authority Intelligence (ARE)

- Core Question: Who holds the legal authority to sign the purchase agreement?
- Mechanics: Evaluates court dockets through a lifecycle state machine. Identifies Letters Testamentary / Administration, Independent Administration powers, and bypasses attorney gatekeepers to isolate direct fiduciary records.
- Proprietary Metric: Four-Tier Authority Classification (Confirmed, Likely Decision Maker, Stakeholder Requiring Verification, Unresolved).



Layer 5: Opportunity Intelligence (OSE)

Opportunity Intelligence converts ownership, control, authority, equity, and friction into a prioritized acquisition recommendation.

- Core Question: What is the probability-adjusted value and outreach priority?
- Mechanics: Calibrates gross equity upside against the Deal Friction Score (DFS). Distinguishes high-equity/low-friction quick closes from high-equity/extreme-friction legal traps.
- Proprietary Output: Composite Score (0–100), Priority Band (Priority A, B, C, Disqualified), and Dynamic Milestone Re-Scoring.

5. Why Probate is Only the First Vertical

Probate is the crucible of real estate title research. It represents the highest data friction, the most fragmented local court systems, and the most intricate legal rules in the property sector. If a platform can solve county-level probate data, it has solved the hardest data problem in real estate. The five-layer architecture of Gieni is completely asset-agnostic and transaction-agnostic.

6. Future Expansion Verticals

- **1. Trust Estate Dispositions:** Uncovering off-market trust-held properties, locating successor trustees, and facilitating private non-court liquidations.
- **2. Heir Property & Clouded Title:** Multi-generational unprobated properties, providing structured genealogical heir mapping and title cures.
- **3. Judicial Partition Actions:** Identifying fractured co-tenancies in active partition lawsuits, enabling investors to acquire fractional interests or broker unified buyouts prior to auction.
- **4. Guardianship & Conservatorship:** Navigating court-supervised liquidations for incapacitated estate owners requiring liquidity for skilled nursing care.
- **5. Tax Distress & Surplus Equity:** Intervening during statutory redemption windows to purchase high-equity distressed homes prior to tax foreclosure.

7. The Intelligence Flywheel (Compounding Scale)



▼
Better Intelligence Models



▼
Better Prioritization



▼
More Customer Success



▼
More Counties

8. What Gieni Produces



Input → Platform → Output

- Inputs: Court Records, Property Records, Tax Records, GIS Records.
- The Core Output: Probate Opportunity File
 - Property Profile (APN, Situs Address, Legal Description, Condition Signals)
 - Ownership Profile (Title Vesting, Deed Chain, Encumbrances & Net Equity Waterfall)
 - Control Profile (De-Facto Decision-Maker, Heir Dynamic, Control Archetype)
 - Authority Profile (Letters Status, Statutory Powers Granted, Court Oversight Requirements)
 - Opportunity Profile (Viability Score 0–100, Priority A/B/C, Deal Friction Score)
 - Evidence Package (Verified Petitions, Deeds, Assessor Cards with SHA-256 Hashes)
 - Recommended Action Directive (First-touch empathy angles, negotiation posture, closing pathway)



Gieni converts fragmented public records into acquisition-ready decisions.